Designing Return Policy in E-Commerce: A Trade-off between Purchase and Return Efficiency

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Abstract

We study the effect of return policies by asking whether they encourage efficient returns and purchases. A stricter return policy curbs inefficient ex-post returns, but the resulting lower return rates make higher product prices more attractive for sellers, leading to possible exclusion of ex-ante efficient purchases. We compare automatic coverage that reimburses return-shipping costs (RI) with optional coverage (CI) by estimating a structural model of purchase, insurance, and return decisions under monopoly pricing across six apparel categories. A lenient return policy like RI performs better in high-margin, high-salvage categories by expanding valuable purchases, whereas a strict return policy like CI performs better in thinner-margin categories by curbing excessive returns. Partial RPD, a menu pairing a discounted price with a medium restocking fee as a penalty, performs better than RI and CI for most categories by better balancing purchase and return efficiency.

Keywords: structural modeling, product returns, return policy, welfare analysis, online retailing.

1 Introduction

Product returns are a central challenge in online retail. Return rates have reached 19.3% in the United States and exceed 35% in China.[1] In monetary terms, total retail returns in the United States are projected to reach $849.9 billion in 2025. The problem is especially acute for apparel, whose fit and quality are difficult to assess online. Industry benchmarks place online apparel return rates in the United States between 20%

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[1]National Retail Federation (2025), 2025 Retail Returns Landscape; Wu Dan Li (2024), Zhejiang University School of Management.

and 40%.[2] These high rates arise in part because consumers must purchase before inspecting the product. Match value uncertainty is resolved only after delivery, when consumers learn their realized match value and decide whether to keep the product. Under a typical return policy, consumers are refunded the product price but bear the cost of return shipping, while the seller incurs handling costs and recovers only part of the product's value. Because the consumer does not bear the seller's handling cost, returns can exceed the efficient level conditional on purchase. When the expected value from keeping is not enough to offset that loss, the purchase itself can generate negative expected social surplus and be ex-ante inefficient.

By considering the efficiency of purchases and returns jointly, the platform can design a return policy to improve the joint welfare of consumers and sellers. A purchase should occur if and only if it creates positive expected social surplus. Once the consumer's match value is realized, the product should be returned if and only if returning creates more social value than keeping. From the welfare perspective, the existing return literature mostly asks whether the consumer returns efficiently given the refund (Su, 2009), but not whether a purchase should occur in the first place.[3] A return policy changes consumers' return behavior, which in turn affects whether a purchase should occur, so purchases and returns are linked. Only by evaluating purchases and returns jointly can we assess the overall efficiency of return policies and determine whether they trade off or simultaneously approach the efficient level.

Return Shipping Insurance (RSI), which is commonly used on Chinese e-commerce platforms, changes both purchases and returns by reimbursing the buyer's return-shipping expense. We examine two RSI policies, automatic coverage for all buyers (*Retailers' Return Shipping Insurance*, RI) and optional coverage at checkout (*Consumers' Return Shipping Insurance*, CI). Automatic coverage encourages inefficient returns with positive consumer benefit but negative social cost. However, it can encourage purchases that create positive value and would not occur if consumers feared paying for return shipping. On the other hand, optional coverage disciplines returns by limiting reimbursement to those who buy insurance, but consumers with a high chance of returning may be especially likely to select into it. A seller's profit from a sale depends on whether the product is ultimately kept or returned. By changing both consumers' willingness to buy and their probability of returning, each insurance arrangement changes expected profit at any given price and

[2] Eightx (2026), Average eCommerce Return Rate by Category; McKinsey & Company (2020), Returning to order.

[3] In Che (1996), consumers are homogeneous and all of them buy. Allowing returns can make the seller charge a higher price. The consumer keeps if and only if valuation exceeds that price. When retail cost is small, that price exceeds cost. The consumer then returns even though keeping is efficient. That loss is a suboptimal return. In a cheap-talk setting, a refund above salvage makes seller advice credible (Inderst and Ottaviani, 2013). This produces overpurchase in the seller's recommendation given the signal. It also produces over-return. The question is whether the seller should send that signal. Buyers are not screened by expected social value.

therefore the price the seller chooses. Once the seller adjusts the price, it is not obvious which arrangement creates more value, or whether that ranking changes with product margins and return costs.

These observations lead to three research questions. First, how do automatic and optional return-shipping insurance affect purchases, insurance selection, returns, and seller pricing? Second, is automatic coverage superior to optional insurance, and how does the result vary across product categories? How much of the difference comes from purchases, and how much from returns? Third, can alternative policies that better balance purchases and returns create greater social value than the existing insurance policies?

In our model, a monopolistic seller sets the product price, anticipating consumers' purchase and return decisions. Consumers only know their expected value of a product when deciding whether to buy and learn their final match value after delivery, when they decide whether to keep the item. Under either policy, consumers receive a full refund of the product price upon return, and insurance reimburses return-shipping fees. The seller pays production cost on every sale, earns the price when the product is kept, and bears net return-handling costs when it is returned.

Our model generates direct behavioral effects and an indirect effect through the equilibrium price response. At a given product price, RI raises willingness to purchase more broadly because every buyer receives coverage without paying a separate premium, but it also lowers the return cost of every purchaser. Under CI, the premium limits coverage and participation, but uninsured purchasers bear the return shipping costs. Equilibrium price effects are ambiguous because three forces interact: the seller-paid RI premium creates cost pass-through, the price itself changes the total volume of returns through who purchases and whether those purchases are returned, and the number of purchases ultimately kept affects the profitability of raising the price. The last force is *surplus extraction*. When the composition of demand produces fewer returns, net demand is higher and a price increase becomes more profitable.

To assess these effects jointly, we consider the welfare consequences of purchases and returns along the ex-ante and ex-post margins. *Ex-ante efficiency* concerns whether purchases create positive expected social surplus. Broader participation under RI can mitigate monopoly underprovision when marginal buyers are socially valuable, but it can also create overprovision by attracting frequent returners whose purchases impose more cost than value. *Ex-post efficiency* concerns whether a purchased product is returned only when doing so creates social value after accounting for shipping, hassle, handling, and recovery value. CI directly strengthens return discipline among uninsured buyers, although its aggregate ex-post advantage also depends on equilibrium prices and insurance take-up. Return policy must therefore preserve valuable purchases while

discouraging both inefficient purchases and excessive returns. RI tends to perform better when margins are high and return-handling costs are low, where the additional purchases from a lower price due to weaker surplus extraction outweigh the return risk. In contrast, CI tends to perform better when margins are thin and return-handling costs are high, so curbing excessive returns outweighs the purchases lost from a higher price due to stronger surplus extraction.

Despite these differences, both RI and CI share design limitations on the ex-ante and ex-post efficiency margins. In addition to aggravating ex-post over-return among insured buyers, subsidizing returns can attract consumers who are especially likely to return, creating adverse selection that worsens the composition of purchases. Furthermore, under CI, standardized insurance premiums are set separately from the seller's product price and need not reflect product-level margins, return-handling costs, or recovery values.

The third research question therefore considers a counterfactual price-refund menu that we call *return-penalty discount (RPD)*. Instead of subsidizing return shipping while keeping the product price and refund rule unchanged, RPD preserves a full-price option with full refund rights and adds a discounted option with a stricter refund. The platform sets the return penalty, while the seller chooses the list price and discount using product-level information on margins and return costs. Consumers who expect to keep the product may select the discount, improving purchaser composition and return discipline. Nevertheless, RPD is not necessarily more efficient. Stricter refund terms can strengthen surplus extraction and exclude socially valuable buyers. If the penalty is too severe, it can also deter returns that would be socially efficient. We evaluate both full forfeiture of the refund and partial forfeiture such as restocking fees. Under monopoly, preserving valuable purchases, discouraging inefficient purchases, and maintaining return discipline remain in tension, so which force dominates is an empirical question.

We study these tradeoffs with transaction data from a mid-sized women's clothing retailer on Taobao that repeatedly switched between RI and CI during 2017–2023. Within-retailer switches provide policy variation that is not confounded by retailer identity. For products sold under both policies, RI is associated with lower prices and higher return rates. Under CI, insured transactions return much more often than uninsured ones, consistent with selection into coverage and a behavioral response to reimbursement. Price comparisons vary with product composition and aggregation, which motivates a framework in which prices, purchases, insurance, and returns adjust jointly.

We estimate a structural model of these decisions under monopoly pricing for six apparel categories. The estimates recover category-specific purchase values, match value uncertainty, and return responses, which

we combine with seller costs and recovery values to evaluate equilibrium welfare. The model reproduces the main qualitative differences between the policies. It underpredicts purchases and widens the conditional RI-CI return gap relative to the data, so we use it primarily for within-category policy rankings. As a benchmark, we make returns socially efficient while retaining monopoly pricing. Even then, about 25%–30% of first-best welfare remains unrealized because prices exclude efficient purchases. Making returns efficient does not eliminate the participation distortion.

The comparison between RI and CI shows how the value of participation relative to return losses determines policy rankings. In the high to mid margin categories, there is a genuine trade-off between participation and returns. For dresses, where margins are high and scrap recovery constitutes a large share of the retail price, RI's broader participation includes socially valuable marginal buyers excluded under CI and outweighs its extra return distortion. In medium-margin categories (outerwear, pants, sweaters, and two-piece sets), CI excludes some efficient purchases, but by curbing inefficient entry and restoring return discipline, it outperforms RI by about 15–30 percentage points of first-best welfare. The starkest contrast is tops, the thin-margin category, where CI outperforms RI on both the ex-ante and ex-post margins. Roughly 80% of first-best welfare is lost under RI through severe ex-post over-return and poor ex-ante screening, while CI attains about 50 percentage points more of first-best welfare.

Full-forfeiture RPD shows why stronger return discipline is not enough by itself. In most categories, purchases concentrate on the discounted branch, full-price demand is negligible, and return probability on the discount branch is near zero. The policy therefore replaces over-return under RSI with under-return, deterring returns that would be socially efficient. Even so, it substantially reduces ex-post distortion. It also strengthens surplus extraction and raises the purchase cutoff, excluding marginal consumers, even though the average price paid is lower. The reduction in return distortion dominates this participation loss in every category except dresses. Outside dresses, full-forfeiture RPD outperforms RI by more than 15 percentage points of first-best welfare and slightly outperforms CI by up to about 7 percentage points, whereas for dresses it falls short of both RSI policies by roughly 17–20 percentage points because many excluded buyers would have created positive social surplus.

Partial-forfeiture RPD softens both failures of full forfeiture. Restocking fees of 10%–15% of the discounted price preserve much of the return discipline while reducing under-return and restoring participation. These fees outperform both RSI policies outside the dress category, attaining about 75–78% of first-best welfare. For dresses, partial forfeiture restores much of what full forfeiture loses (raising welfare from about

57% of first-best under full forfeiture to about 69–70%) but remains below RI (about 76%). For tops, the thin-margin category, full forfeiture slightly outperforms partial-forfeiture RPD in this range. Partial-forfeiture RPD's advantage outside the dress category persists when recovery varies, but RI's baseline advantage for dresses disappears when recovery is low. Resampling largely supports these patterns. The evidence favors category-specific penalties, providing a more generous return policy when margins are high and returns are not socially costly to handle. When tailoring is infeasible, moderate partial-forfeiture RPD may serve as a practical alternative.

1.1 Literature Review

Many existing empirical studies of return policy examine, from the seller's perspective, a trade-off between stimulating purchase and managing return cost. For purchases, return leniency raises current purchases (Wood, 2001), and a completed return raises later demand and lifetime value (Petersen and Kumar, 2009, 2015; Bower and Maxham, 2012; Griffis et al., 2012). For costs, extra reverse-logistics cost can exceed extra orders (Patel et al., 2021), profit falls when the seller pays return shipping (Hjort and Lantz, 2016), and free shipping promotions raise returns enough to cut profit (Shehu et al., 2020). A meta-analysis finds that different return-policy dimensions, such as time or scope, affect purchases and returns differently (Janakiraman et al., 2016). Although this literature generally treats additional purchases as beneficial to the seller, they can reduce the joint surplus of consumers, the seller, and the insurer when return probabilities are high and margins are thin. Under RI, such overprovision can arise because the seller does not fully internalize insurer claim losses. To our knowledge, no empirical study asks whether return policy causes under- or overprovision or compares this distortion across policies. We use two measures. Ex-ante inefficiency is the surplus lost from distorted participation relative to first best when returns are efficient. The purchase-efficient gap is the actual purchase cutoff minus the zero-surplus cutoff under actual return behavior. Positive values indicate underprovision and negative values overprovision. By this measure, CI and RPD underprovide in every category, whereas RI underprovides only for high-margin dresses and overprovides elsewhere.

Price determines who participates, and greater return leniency can raise or lower the equilibrium price. A higher refund raises willingness to pay and allows the seller to extract option value (Su, 2009). A money-back guarantee can also support a higher price by screening keepers (Che, 1996). Allowing returns can make demand less elastic and raise prices (Newberry and Zhou, 2024). More generally, sellers may pass return-related costs through to consumers in product prices, including the seller-paid insurance premium

under RI (Ren et al., 2021; Li et al., 2023b).[4] Conversely, a full refund may induce the seller to cut price to deter trial and return (Davis et al., 1995). One channel missing from the return literature is the net-demand channel for surplus extraction. In our setting, a more generous policy can thin net demand, weakening surplus extraction and putting downward pressure on price.

On the ex-post margin, returns need not be socially wasteful. Su (2009) shows that when the refund equals salvage, a return reallocates the good to the party with the higher value. Later work explains why the seller's optimal refund departs from this benchmark. The refund exceeds salvage when the seller compresses information rents (Courty and Hao, 2000; Matthews and Persico, 2007) or makes sales talk credible (Inderst and Ottaviani, 2013). Manufacturer buy-backs can correct a retailer's excessively strict refund (Shulman et al., 2010). The refund falls below salvage to deter clearance deferrals (Altug and Aydinliyim, 2016), to discourage returns or extract surplus from returners (Shulman et al., 2009; Shang et al., 2017a), or to deter switching to a rival (Shulman et al., 2011). With opportunistic renters, the optimal refund can lie on either side of salvage depending on rental value, salvage, and sales volume (Altug et al., 2021). These studies explain departures from the socially efficient refund but do not empirically quantify or compare return distortions across policies. We measure that distortion as ex-post inefficiency, the welfare lost when purchasers do not return efficiently, holding fixed who bought. In our counterfactuals, ex-post inefficiency is smallest under RPD, larger under CI, and largest under RI in every category.

Theory on RSI compares RI and CI mainly by seller profit, with some analyses of consumer surplus and social welfare. Li et al. (2023b) distinguish informed buyers, who never return, from uninformed buyers, who return at a fixed rate. RI is more profitable when handling costs are low, whereas CI can benefit the retailer, insurer, and consumers when uninsured buyers anticipate regret. Li et al. (2021) find that RI dominates when the premium is low and the refund equals salvage, while CI may leave more surplus with buyers. Low premiums and high compensation also favor RI in Fan and Chen (2020), although adding RSI to a money-back guarantee need not raise retailer profit (Ren et al., 2021). With quality signaling, RSI can harm both high- and low-quality sellers while benefiting consumers, and higher-rated sellers are more likely to adopt RI (Zhang et al., 2022).[5] We instead evaluate RI and CI empirically in terms of retailer profit, expected consumer surplus, and expected social surplus, showing how their ranking varies with margins and handling costs.

[4] These RI–CI models hold each buyer's return probability fixed.

[5] Other empirical work on RSI is scarce. It includes Chen et al. (2018) on how RSI changes purchase and return intentions.

Return-policy design uses both nonmonetary and monetary penalties (Abdulla et al., 2019). Early models deter opportunistic returns through hassle under a full money-back guarantee (Davis et al., 1998) or nonrefundable shipping charges (Hess et al., 1996). Later models jointly choose prices and partial refunds (Shulman et al., 2009; Hsiao and Chen, 2012) or offer self-selective screening menus (Shang et al., 2017a; Altug et al., 2021; Najafi et al., 2024; Huang and Zhang, 2020). Unlike those screening menus, in which the seller sets both the price and the refund, the seller here chooses the list price and the discount, and the platform sets the refund penalty.[6] More recently, Costello and Bechler (2026) and Fang et al. (2026) examine returnless policies that refund the buyer in full or in part while letting them keep the product. Empirical work on return policy finds a large return option value in catalog apparel (Anderson et al., 2009) but a small willingness-to-pay premium for a full refund in electronics (Shang et al., 2017b). The closest empirical study to ours is Newberry and Zhou (2024), who find that the insurance value of returns accounts for roughly 80% of consumer surplus gains, compared with 20% for quality signaling, which supports abstracting from the signaling channel. They estimate the welfare effects of uniform return policies and show that repricing benefits the seller but harms consumers, because sellers pass return costs through and demand is less elastic. By contrast, we study self-selection under CI and RPD rather than a uniform return policy.

To keep the model parsimonious, we abstract from pre-purchase search (Petrikaitė, 2018; Jerath and Ren, 2024; Janssen and Williams, 2024; Li et al., 2024; Ibragimov et al., 2026), purchase-return interdependence over customer lifetimes (Chen, 2021; Montaguti et al., 2026), quality signaling (Moorthy and Srinivasan, 1995; Zhang et al., 2022; Li et al., 2023a), inter-firm competition over return policy, such as money-back guarantees, restocking fees, or RI (Chesnokova, 2007; McWilliams, 2012; Shulman et al., 2011; Chen and Grewal, 2013; Inderst and Tirosh, 2015), and supply chain buyback contracts (Padmanabhan and Png, 1997; Tran et al., 2018).

Our main contribution is an empirical welfare decomposition that separates losses from inefficient participation, when inefficient purchases occur or efficient purchases are deterred, from losses due to suboptimal return decisions. By comparing these two margins across policies and categories, the framework identifies when they can improve together and when they trade off. In the latter case, we show that whether to prioritize participation or return discipline depends on margins and return-handling costs. Applied to RSI and RPD, the framework provides, to our knowledge, the first empirical evaluation in online retail of both RSI and a

[6] In practice, refund amounts are often constrained by platform rules or law. Amazon permits merchant-fulfilled sellers to charge a restocking fee of up to 20% for change-of-mind returns received after the return window (Amazon Seller Central Help). Taobao and JD.com instead implement China's seven-day no-reason return with a full product-price refund.

self-selective price-refund menu, measuring seller profit, expected consumer surplus, and expected social surplus.[7] These results can guide platforms on when to tighten or relax return policy and support a moderate restocking fee in that menu as a practical alternative across most apparel categories when tailoring requires information the platform lacks.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting of return shipping insurance, introduces the transaction data, and documents key stylized facts on prices and returns. Section 3 sets up the structural model, and Section 4 characterizes equilibrium purchase, insurance, and return decisions, seller pricing incentives, and the welfare framework. Sections 5 and 6 discuss identification, estimation, and calibration. Section 7 reports counterfactual welfare simulations comparing RI, CI, and return-penalty discount (RPD) by category. Section 8 concludes.

2 Institutional Setting and Data

Apparel accounts for the largest share of online retail in China at 22%.[8] Because consumers cannot inspect garments before delivery, return rates are high, historically generating frequent and costly return-shipping disputes.

2.1 Return Shipping Insurance: RI versus CI

To resolve frequent and costly return-shipping disputes, Taobao and Huatai Insurance introduced Return Shipping Insurance (RSI) in 2010 to standardize reimbursement. The RSI market reached approximately 8 billion RMB in 2022.[9] The insurance market on Taobao is centralized. The platform partners with a small set of insurers and assigns providers algorithmically, so coverage terms are uniform across merchants. On approved returns, buyers receive a full refund of the product purchase price from the seller and reimbursement of the 1 kg delivery fee from the insurer (typically 5 to 25 RMB).

RSI operates under two mutually exclusive regimes. Under RI, the seller enrolls at the store level, without a separate consumer fee. The insurer charges the seller a per-transaction premium (typically 1 to 5 RMB) based on the store's three-month historical return rate and category benchmarks. RI dominates the platform, accounting for over 90% of all RSI policies. Under CI, which is offered *if and only if* the seller has not enrolled the store in RI, individual buyers choose whether to purchase coverage as an add-on at checkout. In

[7] Vollmer (2025) structurally compares resale with a single partial refund in college football tickets, reporting primary-seller profit, consumer welfare, and total welfare. The seller posts one refund per game, and a holder returns if realized value falls below it. Buyers do not choose among refund contracts, and the refund is not paired with an upfront discount.

[8] Ministry of Commerce (2024), *2023 China Online Retail Market Development Report*.

[9] Shanghai Securities News (2022), Sina Finance.

contrast to RI, the CI premium is consumer-specific, determined mainly by the buyer's province of residence and return history, *independent of the product*. CI premiums typically exceed RI premiums because optional coverage attracts consumers who anticipate returning, and insurers charge higher rates to buyers with higher return propensities.

2.2 Data and Sample Construction

Our dataset comprises transaction records from a single mid-sized women's clothing retailer on Taobao, spanning from 2017 to 2023. Each transaction falls under one RSI policy. Over this period, total revenue is approximately 62 million RMB.

The data include sales, returns, and insurance at the transaction level, with consumer information limited to addresses. Insurance premium and coverage are observed for nearly all RI transactions, excluding international orders and buyers ineligible because of high historical return rates. Under CI, coverage is recorded only when the consumer buys insurance, while for non-purchasers we impute it from same-province addresses. Premiums are not recorded, so we impute the premium from coverage using the 2025 premium-coverage schedule (Online Appendix A).

We code a return as a binary indicator equal to one when the buyer receives a strictly positive refund and zero otherwise, excluding partial refunds for minor product defects. We keep single-item, single-quantity orders with confirmed shipment and no premium shipping, and drop non-product listings and promotion days on which the store temporarily switched RSI policy. Restricting to one item per order rules out complementarities between purchase and return decisions across products, while restricting to quantity one rules out bracketing, in which buyers order multiple sizes or colors of the same listing and return all but one. We keep six apparel categories with at least 50 transactions per month on average: dress, tops, pants, sweater, two-piece, and outerwear. The cleaned sample contains 361,201 transactions (238,612 under RI and 122,589 under CI).

2.3 Stylized Facts

Figure 1 plots 30-day moving averages of the daily return rate from 2017 through 2023 across RI and CI periods. Over these six years, the store repeatedly switched between the two policies. These switches do not seem to be based on the return rate. They occur when returns are low or high and when the rate is rising or falling. Separately, the return series drifts upward over the sample and exceeds 40% toward the end of 2023 during a long period of RI. This late-sample climb plausibly coincides with the rapid expansion of

live-stream commerce on Taobao, where purchases are typically more impulsive and return-prone.[10]

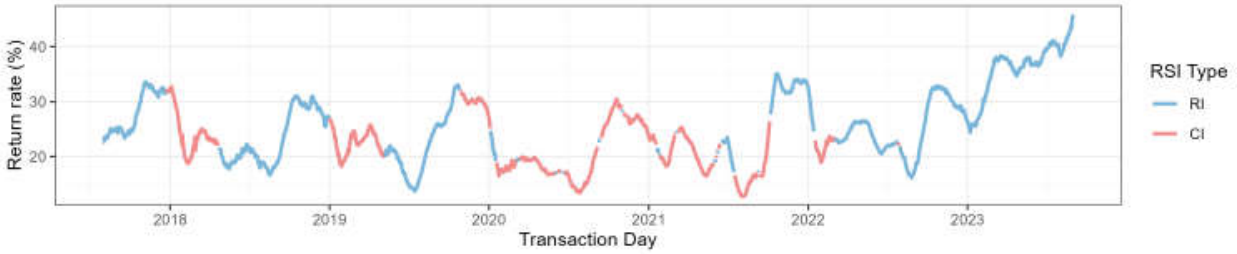


Figure 1: 30-day moving average of the daily return rate (%), 2017–2023, with RI and CI periods shaded.

Table 1 reports prices, premiums, coverage, and return rates under RI and under CI, pooling across categories.

Table 1: Summary Statistics

RSI	Variable	Unit	N	Mean	Sd	Min	Median	Max
CI	Price	RMB	122,589	91.50	56.67	18.0	83.65	1038.0
	Premium	RMB	122,589	5.01	0.22	4.6	5.10	7.1
	Coverage	RMB	122,589	10.63	1.46	8.0	11.00	25.0
	Return Rate	%	122,589	21.10	40.80	0.0	0.00	100.0
RI	Price	RMB	238,612	87.36	67.58	15.3	79.00	738.0
	Premium	RMB	238,612	2.67	0.39	1.4	2.50	3.3
	Coverage	RMB	238,612	10.59	1.45	8.0	11.00	25.0
	Return Rate	%	238,612	27.30	44.55	0.0	0.00	100.0

Notes: Return-rate medians are zero because the outcome is binary.

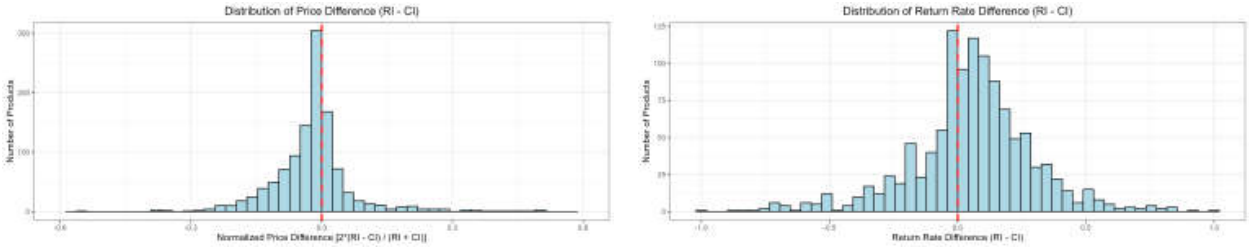
Average list prices are higher under CI than under RI (about 92 versus about 87 RMB). Traditional premium pass-through would predict the opposite. The seller-paid RI premium is a visible cost that could raise list prices under RI. We do not observe higher RI prices, suggesting that other pricing forces operate. One candidate force is surplus extraction. Under CI, uninsured buyers face higher private return costs and return less often, which can strengthen the seller's incentive to set higher prices.

Average return rates are higher under RI than under CI, with pooled means of about 27% versus about 21%. Under CI, insurance is purchased on 2.91% of transactions. Return rates are much higher among insured than uninsured transactions (46.3% versus 20.4%). The gap is consistent with lower private return costs among insured buyers and adverse selection into coverage.

Pooled comparisons may be confounded by product selection. Figure 2 restricts the sample to 1,140 products that sold under both RI and CI and have more than five transactions per product. Panel (a) plots the normalized price difference $2(\bar{p}^{\mathrm{RI}} - \bar{p}^{\mathrm{CI}})/(\bar{p}^{\mathrm{RI}} + \bar{p}^{\mathrm{CI}})$, where $\bar{p}$ is the product-level mean transaction price under each policy. As in the pooled sample, a larger share of products still show higher prices under CI than under RI. Panel (b) reports $\bar{r}^{\mathrm{RI}} - \bar{r}^{\mathrm{CI}}$, where $\bar{r}$ is the product-level mean return rate under each policy. Most

[10] Removing a pooled annual cycle (day-of-year harmonics 1–4) prior to the same 30-day moving average yields a broadly similar pattern, including the pronounced late-sample increase, and still shows no clear ranking of return rates between RI and CI periods.

mass lies to the right of zero, indicating higher return rates under RI.



(a) Normalized price difference. (b) Return-rate difference.

Figure 2: Within-product price and return-rate differences (RI − CI).

Taken together, the time series documents repeated within-store policy switches. Pooled and within-product comparisons show that a larger share of products have higher prices under CI than under RI, and that return rates are higher under RI once products are held fixed.

3 Model

We consider a retail shopping model with RSI in which a monopolistic seller offers apparel category j in period t. The seller faces downward-sloping residual demand, with the outside option absorbing purchases from rivals. Market size N_{jt} varies across categories and seasons (Online Appendix A). Categories are separable, with no cross-category demand spillovers. For notational simplicity, we suppress the category subscript j and period subscript t throughout Sections 3 and 4, reintroducing them in Section 5 for identification across categories and markets.

The game proceeds in three stages. First, the seller sets the retail price p, taking the RSI policy and insurance premiums as given. Second, each consumer decides whether to purchase the product and, under CI, whether to purchase optional insurance. Third, after delivery, consumers observe their idiosyncratic match value shocks and choose whether to keep or return the product.[11]

The insurer provides consumer-specific coverage k_i according to the location-specific 1 kg delivery fee. Under RI, the seller pays premium I^r per transaction, and coverage is automatic for all buyers. Under CI, buyers choose whether to purchase coverage at checkout for premium $I^c(k_i)$.

The seller produces each unit at marginal cost $\theta^c = X\alpha^c + \zeta^c$. The store-level shock ζ^c is common across products and reflects shared inputs such as marketing, customer acquisition and service, labor, utilities, and rent. If a unit is returned, the seller incurs a handling fee and recovers resale value, yielding net return handling cost h. This cost is negative when recovery exceeds the handling cost.

[11] Hong and Pavlou (2014) find that, for experience goods, returns are driven more by match value uncertainty than by quality uncertainty.

Consumers differ in ex-ante valuation ν_i and location-specific return-shipping coverage k_i. Ex-ante valuation is

$$\nu_i = \mu + \xi + \omega\epsilon_i, \tag{1}$$

where $\mu = X\beta^\mu$ is mean valuation from observed category attributes X, ξ is an unobserved demand shock orthogonal to X, ω is the standard deviation of idiosyncratic ex-ante tastes within the category and is time-invariant, and $\epsilon_i \sim \mathcal{N}(0,1)$, so $\nu_i \sim \mathcal{N}(\mu + \xi, \omega^2)$ with distribution F_ν. Return-shipping coverage k_i is drawn from a location-specific distribution $\Psi(\cdot)$, independent of ν_i.[12]

After delivery, the idiosyncratic match value shocks of consumer i, $\varepsilon_i^{\text{keep}}$ and $\varepsilon_i^{\text{return}}$, are realized. They follow a Type I extreme value distribution with mean zero and scale $1/\sigma$, with ex-post precision $\sigma = X^\sigma\beta^\sigma$. Conditional on purchase, consumer i chooses between keeping and returning. On a return, the buyer receives a refund of p, incurs hassle κ and shipping k_i, and is reimbursed only for k_i when coverage applies. Realized utility on each branch is the sum of mean indirect utility and the idiosyncratic shock, $u_i = \overline{u}_i + \varepsilon_i$. Mean utilities under the respective policies are

$$\overline{u}_i^{\text{keep}} = \nu_i - p - I^c(k_i)BC_i, \tag{2}$$

$$\overline{u}_i^{\text{return}} = -\kappa - k_i(1 - RSI_i) - I^c(k_i)BC_i, \tag{3}$$

where BC_i is an indicator that consumer i purchases CI coverage, and RSI_i is an indicator of insurance, either under RI or under a CI purchase. Under RI, coverage is automatic, so return shipping is reimbursed up to k_i, yielding $\overline{u}_i^{\text{return}} = -\kappa$. Under CI, an insured buyer receives reimbursement and obtains $\overline{u}_i^{\text{return}} = -\kappa - I^c(k_i)$, whereas an uninsured buyer bears the shipping cost, obtaining $\overline{u}_i^{\text{return}} = -(\kappa + k_i)$.

4 Equilibrium Analysis

We focus on equilibrium within a single category and period. Superscript l denotes the RSI scheme (r for RI, c for CI). We analyze consumer behavior under each RSI scheme and then compare the mechanisms that matter for pricing and welfare.

[12] Coverage depends on location, which may correlate with income. We assume that, conditional on k_i, location does not shift valuation.

4.1 Consumer Behavior under RSI

Under RI, automatic coverage yields mean utilities $\overline{u}_i^{\text{keep}} = \nu_i - p$ and $\overline{u}_i^{\text{return}} = -\kappa$. Integrating over Type I extreme value shocks yields expected consumer surplus

$$CS_i^r = \frac{1}{\sigma} \ln\left(e^{\sigma(\nu_i - p)} + e^{-\sigma\kappa}\right). \tag{4}$$

Consumers purchase if $CS_i^r \geq 0$, or $\nu_i \geq \hat{\nu}^r \equiv p + \frac{1}{\sigma}\ln(1 - e^{-\sigma\kappa})$. Conditional on purchase, the return probability is

$$q_i^r \equiv \frac{e^{-\sigma\kappa}}{e^{\sigma(\nu_i - p)} + e^{-\sigma\kappa}}. \tag{5}$$

Lower precision σ (higher match value uncertainty) raises the option value, increasing both consumer surplus and return probability.

Under CI, consumers choose between purchasing with optional insurance at premium $I^c(k_i)$, purchasing without insurance, and not purchasing. Expected consumer surplus under the two purchase options is

$$CS_i^{bc} = \frac{1}{\sigma} \ln\left(e^{\sigma(\nu_i - p)} + e^{-\sigma\kappa}\right) - I^c(k_i), \qquad CS_i^n = \frac{1}{\sigma} \ln\left(e^{\sigma(\nu_i - p)} + e^{-\sigma(\kappa + k_i)}\right). \tag{6}$$

A purchaser prefers insurance if $CS_i^{bc} \geq CS_i^n$, which holds when valuation is below the insurance-indifference cutoff

$$\nu_i \leq \tilde{\nu}^{bc} \equiv p - \kappa + \frac{1}{\sigma} \ln\left(\frac{e^{-\sigma I^c(k_i)} - e^{-\sigma k_i}}{1 - e^{-\sigma I^c(k_i)}}\right). \tag{7}$$

Because return probability falls with valuation, insurance is valued most by lower-valuation buyers.

Setting $CS_i^{bc} \geq 0$ and $CS_i^n \geq 0$ yields purchase thresholds $\hat{\nu}^{bc} \equiv p + \frac{1}{\sigma}\ln\left(e^{\sigma I^c(k_i)} - e^{-\sigma\kappa}\right)$ and $\hat{\nu}^n \equiv p + \frac{1}{\sigma}\ln\left(1 - e^{-\sigma(\kappa + k_i)}\right)$. These cutoffs characterize optimal choice: consumers purchase with insurance if $\hat{\nu}^{bc} \leq \nu_i < \tilde{\nu}^{bc}$, purchase without insurance if $\nu_i \geq \max\{\tilde{\nu}^{bc}, \hat{\nu}^n\}$, and do not purchase otherwise. An insured segment exists when $\tilde{\nu}^{bc} \geq \hat{\nu}^{bc}$,[13] which holds when coverage k_i is large relative to the premium $I^c(k_i)$ and hassle cost is not too high. If hassle cost is too high, return probability is too low for the option value of insurance to outweigh the premium, so all purchasers remain uninsured.

Insured consumers return with probability q_i^r, as under RI, because the premium is sunk at purchase. Uninsured consumers bear the full return cost k_i and return with lower probability

$$q_i^n \equiv \frac{e^{-\sigma(\kappa + k_i)}}{e^{\sigma(\nu_i - p)} + e^{-\sigma(\kappa + k_i)}}. \tag{8}$$

[13] This is equivalent to $e^{-\sigma\kappa}(1 - e^{-\sigma k_i}) \geq e^{\sigma I^c(k_i)} - 1$.

4.2 Retailer's Decisions

The seller prices against gross demand and returns. Let D_i^l denote the purchase indicator implied by the cutoffs above, and let q_i^l denote the corresponding return probability under scheme $l \in \{r, c\}$ conditional on purchase: q_i^r is given by (5), and q_i^c equals q_i^r for insured purchasers and q_i^n in (8) for uninsured purchasers. Aggregating individual purchase choices and return probabilities over F_ν and Ψ yields gross demand $D^l \equiv N\,\mathbb{E}[D_i^l]$ and return volume $R^l \equiv N\,\mathbb{E}[D_i^l q_i^l]$. Net demand, $D^l - R^l$, is the number of units ultimately kept. Let *RI* denote an indicator equal to one under RI and zero under CI. The seller's aggregate profit is

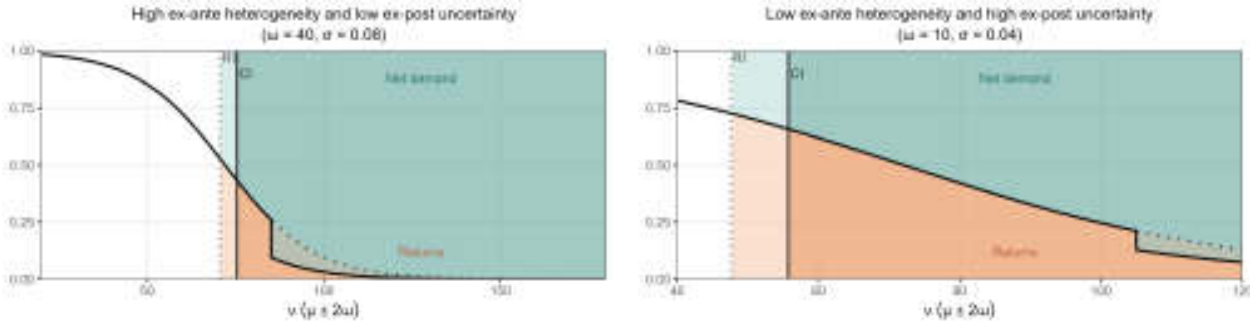
$$\pi^l(p) = (p - \theta^c - I^r RI)(D^l - R^l) - (h + \theta^c + I^r RI)R^l. \tag{9}$$

The first term is the margin earned on kept units, while the second is the loss on returned units. At the same price, RI generates higher gross demand than CI because consumers face no explicit premium, but it also attracts more lower-valuation buyers and reimburses return shipping for all buyers. Both raise return volume. The equilibrium price therefore maximizes (9) against net demand $D^l - R^l$ and return volume. Three channels shape pricing: cost pass-through, surplus extraction, and return mitigation. These forces need not move prices in the same direction.

Cost Pass-through. Under RI, the seller pays the insurance premium I^r, which enters as a marginal cost, so a higher premium raises prices through pass-through. Under CI, the premium does not enter profit directly and affects profit only through demand and returns. Incidence among purchasers also differs. Under CI, consumers pay premiums directly, so insurance costs fall on lower-valuation buyers who return more often. Under RI, premiums are passed through to the product price and borne by all purchasers, so high-valuation keepers effectively subsidize insurance for lower-valuation frequent returners.

Surplus Extraction. Stronger net demand (more keepers) makes a price increase more profitable. We call this force *surplus extraction*. At a common price, RI expands gross demand (no premium) but also raises the return rate (automatic coverage), so net demand need not rise. Figure 3 shows the comparison. The CI purchase cutoff lies to the right of the RI cutoff because gross demand is lower at the same price. Under CI, $q(\nu)$ also drops sharply at the insurance cutoff, where private return costs jump.

Whether net demand is higher under RI or CI depends on ex-ante heterogeneity ω and ex-post uncertainty. When ex-ante heterogeneity is large and ex-post uncertainty is small, as in Figure 3 (left), gross demand is relatively inelastic and return probability is sensitive to reimbursement. Return probability is strongly shifted



(a) Large ω, large σ ($\omega = 40$, $\sigma = 0.08$).

(b) Small ω, small σ ($\omega = 10$, $\sigma = 0.04$).

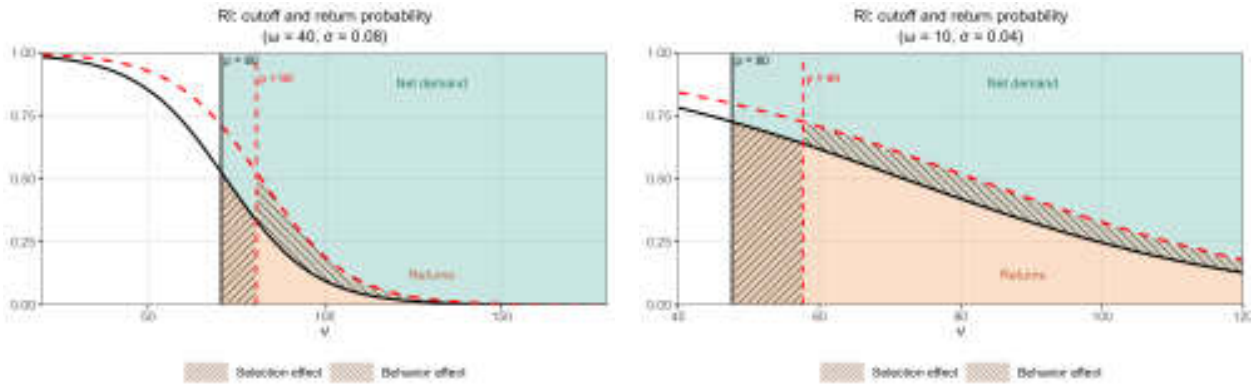
Figure 3: Returns and net demand under RI and CI at a common price. Dotted lines are RI and solid lines are CI. Purchasers lie to the right of each cutoff. Downward-sloping curves are return probabilities $q(v)$. Orange is the return region and teal is net demand. The horizontal axis covers $\mu \pm 2\omega$ with $\mu = 100$.

by reimbursement because mean keep-or-return payoffs dominate idiosyncratic shocks. CI then generates higher net demand because uninsured consumers have substantially lower return rates and the reduction in gross demand from requiring consumers to pay the premium is small. When instead ex-ante heterogeneity is small and ex-post uncertainty is large, as in Figure 3 (right), return probabilities respond less to coverage and the premium shifts the purchase cutoff more compared to the concentrated value distribution, so RI's broader participation can dominate and lead to a higher net demand than CI.

An important pricing implication is that when return probabilities are highly sensitive to coverage, a smaller but better-composed purchaser pool under CI can raise net demand and give the seller an incentive to charge a higher price. The surplus extraction force should not be equated mechanically with a higher equilibrium price, because the observed price also reflects pass-through and return mitigation.

Return Mitigation. A higher price affects return volume through two opposing forces. It screens out lower-valuation buyers, who return more often, which tends to lower return volume (a selection effect). Among remaining purchasers, however, a higher price raises individual return probabilities, because consumers receive a larger refund upon return (a behavior effect). Because these forces pull in opposite directions, the net effect is again ambiguous. Figure 4 illustrates both channels under RI for the same (ω, σ) cases as Figure 3 (CI is similar).

When ex-ante heterogeneity is large and ex-post uncertainty is small (Figure 4, left), $q(v)$ responds strongly to price, so the seller prefers a lower price to curb returns. Cutting price reduces each buyer's return probability without much increase in sales. When instead ex-ante heterogeneity is small and ex-post uncertainty is large (right), purchase volume responds strongly to price and $q(v)$ responds less, so the seller prefers a higher price to curb returns. Raising price mainly discourages purchases. When returns are costly,



(a) Large ω, large σ ($\omega = 40$, $\sigma = 0.08$). (b) Small ω, small σ ($\omega = 10$, $\sigma = 0.04$).

Figure 4: Selection and behavior effects of a price increase under RI. Solid curves are the baseline price. Dashed red curves are a higher price. Striped regions mark selection (between cutoffs) and the behavior effect (the upward shift in $q(v)$ among remaining purchasers). Parameters match Figure 3.

the incentive to mitigate returns is stronger under RI. Whether that incentive raises or lowers the RI price depends on these primitives.

Summary. The three channels imply that RI need not have higher equilibrium prices than CI even though the seller pays the premium. When the premium is low and pass-through is weak, surplus extraction and return mitigation can reverse the ranking. RI prices tend to be higher when ex-ante heterogeneity is small and ex-post uncertainty is large, because RI can expand net demand and return mitigation then works mainly through screening out frequent returners. The opposite ranking arises when ex-ante heterogeneity is large and ex-post uncertainty is small. The price comparison is therefore an equilibrium outcome rather than a direct measure of insurance incidence.

4.3 Welfare Framework

We evaluate return policies using a welfare framework that separates ex-post return distortions from ex-ante market participation distortions.

Ex-Post Efficiency. Conditional on purchase, production cost θ^c is sunk. A return is ex-post efficient if and only if the social value of keeping falls short of the social value of returning:

$$v_i + \varepsilon_i^{\text{keep}} \leq -(\kappa + k_i + h) + \varepsilon_i^{\text{return}}, \tag{10}$$

yielding the socially optimal return probability

$$q_i^*(v_i, k_i; \sigma, \kappa, h) \equiv \frac{e^{-\sigma(\kappa + k_i + h)}}{e^{\sigma v_i} + e^{-\sigma(\kappa + k_i + h)}}. \tag{11}$$

Because product prices and insurance payments are transfers, q_i^* evaluates returns against valuation v_i and internalizes net handling cost h. Under RSI, buyers instead compare $v_i - p$ against private return costs, ignoring the seller's return losses and generating excessive returns ($q_i^l > q_i^*$).

Ex-Ante Efficiency. Ex-ante efficiency concerns the purchase decision. Expected seller profit from consumer i is $\pi_i^l \equiv (1 - q_i^l)(p - \theta^c - I^r RI) - q_i^l(h + \theta^c + I^r RI)$, and insurer profit $\tilde{\pi}_i^l$ equals premiums collected minus expected reimbursements paid. Summing these with consumer surplus, retail prices and insurance transfers cancel out, leaving individual social surplus

$$S_i^l \equiv CS_i^l + \pi_i^l + \tilde{\pi}_i^l. \tag{12}$$

A transaction is ex-ante efficient if and only if $S_i^l \geq 0$. We evaluate welfare per potential consumer and hence abstract from market size N. Non-purchasers receive zero. Expected consumer surplus and social welfare under scheme l are $CS^l \equiv \mathbb{E}[CS_i^l D_i^l]$ and $W^l \equiv \mathbb{E}[S_i^l D_i^l]$, where under CI terms are branch-specific across insured and uninsured purchases.

Welfare Decomposition and Extensive Margin. Because S_i^l evaluates surplus at private return behavior q_i^l, policy comparisons entangle purchase and return distortions. To evaluate them on a common basis, we compare welfare W^l to first best using an intermediate benchmark that holds observed purchase decisions D_i^l fixed but enforces optimal return behavior q_i^*. Substituting q_i^* into individual social surplus yields the closed form

$$S_i^*(v_i, k_i) \equiv -\theta^c + \frac{1}{\sigma} \ln\left(e^{\sigma v_i} + e^{-\sigma(\kappa + k_i + h)}\right). \tag{13}$$

Let $\hat{v}^*(k_i)$ denote the first-best purchase cutoff solving $S_i^*(v_i, k_i) = 0$, and let $D_i^* \equiv \mathbf{1}\big(v_i \geq \hat{v}^*(k_i)\big)$. Defining $W^{OR} \equiv \mathbb{E}[S_i^*(v_i, k_i) D_i^l]$ as welfare with optimal returns at observed purchases and $W^{FB} \equiv \mathbb{E}[S_i^*(v_i, k_i) D_i^*]$ as first-best welfare, the welfare gap decomposes into

$$W^{FB} - W^l = \underbrace{\left(W^{OR} - W^l\right)}_{\Delta^{ex\text{-}post}} + \underbrace{\left(W^{FB} - W^{OR}\right)}_{\Delta^{ex\text{-}ante}}. \tag{14}$$

The first term, $\Delta^{ex\text{-}post}$, measures surplus lost to inefficient returns among purchasers under scheme l, holding D_i^l fixed. The second term, $\Delta^{ex\text{-}ante} = \mathbb{E}[S_i^*(v_i, k_i)(D_i^* - D_i^l)] = W^{FB} - W^{OR} \geq 0$, measures the remaining loss from distorted market participation relative to first best. While this participation gap can theoretically reflect either under- or overprovision, return option value implies that the first-best cutoff is below production cost ($\hat{v}^* < \theta^c$), while the seller's market power pushes the purchase cutoff well above it

($\hat{v}^l > \hat{v}^*$). In our counterfactuals, $\Delta^{ex\text{-}ante}$ therefore always reflects underprovision relative to first best.

However, a limitation of $\Delta^{ex\text{-}ante}$ is that it evaluates entry holding returns at the hypothetical first-best rule q_i^*, obscuring whether admitting marginal buyers under actual, distorted return discipline creates or destroys surplus. To evaluate the policy-relevant welfare effect of participation, we therefore introduce another measure, which we call the *extensive margin*, evaluating individual surplus S_i^l at consumers' actual return behavior q_i^l. We quantify this margin using the purchase-efficient gap, defined as the purchase cutoff $\hat{v}^l$ minus the efficient cutoff, where the efficient cutoff is the valuation that solves $S_i^l(v_i) = 0$. If the gap is positive, the seller underprovides by excluding buyers with $S_i^l \geq 0$, and if it is negative, the seller overprovides by admitting buyers with $S_i^l < 0$. Section 7 applies this framework at estimated parameters to evaluate monopoly second-best pricing, RI versus CI, and counterfactual RPD menus.

5 Identification

The primitives to be identified are the coverage distribution $\Psi(\cdot)$, category-specific taste dispersion ω_j, mean ex-ante value μ_t, ex-post precision σ_t, and marginal production cost θ_t^c. Net return handling cost h_t is calibrated in Section 6. As in Anderson et al. (2009), high ex-post uncertainty and low hassle cost both increase purchases and returns. We normalize hassle cost to 5 RMB so that ex-post uncertainty can be separated from hassle cost.[14]

Under RI, fully reimbursed shipping makes the purchase threshold independent of k_i, so purchase does not select on coverage and $\Psi(\cdot)$ is non-parametrically identified from the empirical coverage distribution.

Let $\bar{w}_t^r$ be mean willingness to pay. Under RI, the purchase probability is $\mathbb{E}[D_{it}^r(k_i, p_t)] = 1 - \Phi\big(\frac{p_t - \bar{w}_t^r}{\omega_j}\big)$. Given a price instrument, variation in p_t across category-months traces out this purchase schedule. How fast the purchase probability falls in price identifies taste dispersion ω_j, and where the schedule lies identifies mean willingness to pay $\bar{w}_t^r$. $\bar{w}_t^r$ decomposes into intrinsic value and return option value:

$$\bar{w}_t^r \equiv \underbrace{\mu_t + \xi_t}_{\text{Intrinsic value}} + \underbrace{\left[-\frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t \kappa})\right]}_{\text{Return option value}}.$$

While purchase variation identifies $\bar{w}_t^r$, it cannot separate intrinsic value from return option value. Expected returns provide the additional identification that separates them. Holding purchase probability (and thus $\bar{w}_t^r$) fixed, expected returns strictly increase in match value uncertainty, so joint variation in purchases and returns across markets identifies σ_t, which backs out the option value and isolates intrinsic

[14] Normalizing $\kappa = 0$ would imply that, under RI, the purchase probability is always one.

value $\mu_t + \xi_t$ (Online Appendix B). Variation in observable characteristics X_t then separates mean utility $\mu_t = X_t\beta^\mu$ from demand shocks ξ_t. CI purchase, return, and insurance moments further restrict the same demand primitives through endogenous insurance selection under CI.

Finally, given calibrated net handling cost h_t, inverting the retailer's pricing first-order condition at observed prices maps estimated demand responses into marginal cost θ_t^c. Cross-market variation in cost shifters identifies cost parameters α^c and store-level cost shocks ζ_t^c via $\theta_t^c = X_t\alpha^c + \zeta_t^c$.

6 Estimation

We estimate the model by GMM, pooling moment conditions across categories j. The estimation sample contains 6 apparel categories and 313 category × month markets after smoothing, resampling, and the 5% purchase-probability screen described in Online Appendix A. Smoothing and resampling capture stable long-run demand. The screen keeps purchase probabilities away from zero to avoid the zero-share problem.[15] Market size is defined at the category × season level as twice the maximum monthly sales within that category-season, and monthly purchase probabilities equal smoothed sales divided by that size. Because the outside option is unobserved, absolute market size cannot be separately identified from the mean utility level. Setting market size to twice peak sales keeps purchase probabilities interior and comparable across seasons. Category-specific taste dispersion ω_j is fixed over time.

Each category-month market has a single list price and a single net handling cost, including months in which RI and CI both appear. Purchase, return, and insurance-purchase counts under each scheme are scaled by the time share of that scheme, so each scheme is treated as if it had operated for the full month at that common price. That price balances the seller's pricing incentives under both schemes in that category-month.

6.1 Return-handling calibration

Net return handling cost is h_t = gross handling − scrap value$_t$. We set gross handling to 15 RMB per return, consistent with industry reports indicating that each apparel return costs sellers approximately 15 RMB in sunk marketing, packaging, and reverse logistics.[16] For scrap value, we calibrate a three-way disposition of returned inventory using industry surveys. About one-half of returned apparel units are resold at full retail

[15] A low purchase probability makes the purchase-probability inversion numerically unstable.

[16] China Venture News and Zhou Jiabao (2025), reprinting *Times Finance*, cite Yinman founder Fang Jianhua on per-return seller costs for Chinese e-commerce apparel. Yinman operates reverse-logistics services for online retailers; the 15 RMB figure is described as the cost of a return that generates no resale revenue, net of any recovery. We treat this as a per-unit processing charge (reverse shipping, inspection, repackaging, and a share of customer-acquisition and fulfillment costs allocated to the return event) and hold it fixed across categories and months.

price (e.g., 48%). Condition grading assigns roughly 20–30% of units to discount or outlet channels and 20–30% to secondary liquidation; we combine these into a 40% markdown tail. Roughly 10% of returns are written off.[17] Weighting across these channels yields an average scrap recovery of $0.5\,p_t + 0.4 \times (0.5\text{–}0.7)\,p_t \approx 0.75\,p_t$, implying baseline net handling cost $h_t = 15 - 0.75\,p_t$ (RMB). Online Appendix D reports robustness checks under alternative scrap shares of $0.65\,p_t$ and $0.85\,p_t$.

6.2 Demand Side

We estimate demand parameters following the nested fixed-point approach of Berry et al. (1995). The nonlinear parameters $\Theta = (\beta^\omega, \beta^\sigma)$ determine ex-ante valuation dispersion, $\omega_j = C_j'\beta^\omega$ for a category-indicator vector C_j, and ex-post precision, $\sigma_t = X_t^\sigma \beta^\sigma$ for a market-covariate vector X_t^σ. Mean utility is $\mu_t = X_t\beta^\mu$, where X_t contains category-month demand shifters: assortment size, in-season share, newly published share, category indicators, and garment style attributes (length, cut, and material texture). Because β^σ is identified from cross-market variation in σ_t, we keep X_t^σ parsimonious, with only category indicators and newly published share. Assortment size and in-season share shift ex-ante value and procurement cost (μ_t and θ_t^c) rather than ex-post uncertainty, and are therefore excluded. We normalize $\kappa = 5$ RMB and fix $\Psi(\cdot)$ at the empirical RI coverage distribution.

For each candidate Θ, we recover the mean-utility index $\delta_t(\Theta) = \mu_t + \xi_t$ by inverting model purchase probabilities. In single-policy markets, the inversion matches observed coverage-weighted purchase probabilities. When RI and CI coexist within a market, a single δ_t minimizes the sum of squared discrepancies in coverage-weighted purchase probabilities across both policies. We estimate $\beta^\mu(\Theta)$ by IV regression of $\delta_t(\Theta)$ on X_t and define structural demand shocks $\xi_t(\Theta) = \delta_t(\Theta) - X_t\beta^\mu(\Theta)$.

The instrument vector Z_t includes X_t and a Hausman-style cost instrument: the contemporaneous average price of other apparel categories sold by the retailer. Common input costs (textiles, labor, warehousing, and platform logistics) induce joint price variation across categories. Normalizing purchase probabilities by category-season market size removes the common seasonal demand shock across categories, and averaging prices within a category lessens the demand-shock links between individual garments.[18] With these

[17] 48% full-price resale: Eightx (2026, Average eCommerce Return Rate by Category), a U.S. e-commerce apparel benchmark. SELVANE (2024, The hidden cost of fast fashion returns) sorts returns into full-price, discount or outlet (about 20–30%), and secondary liquidation (about 20–30%), with 25–30% never returning at list price; we map the last two channels into the 40% markdown tail. McKinsey & Company (2020, Returning to order) estimates that about 10% of apparel returns are discarded; remaining units that fail full-price resale enter the markdown tail and recover 50–70% of list price. The same shares apply in every category, so h_t varies only through p_t.

[18] A pair of pants may go especially well with a certain shirt, which links the demand shocks for those two garments. Averaging prices within each category lessens those links. Market size equals twice peak monthly sales in the category-season and is held fixed

adjustments, category-specific demand shocks ξ_{jt} remain largely orthogonal to demand shocks in other categories, so other-category prices pick up little own-category demand. The first-stage partial F-statistic of this instrument for own price is 100.07 ($p = 0.000$; $N = 313$), well above conventional thresholds for weak instruments.

Given candidate Θ, we form two sets of GMM moments: orthogonality conditions $E[Z_t \xi_t(\Theta)] = 0$ and squared prediction errors $m_t(\Theta)^2$ matching sample return and insurance rates to their model-implied counterparts, where $m_t(\Theta)$ is the coverage-weighted gap between sample and implied rates.[19] We choose $\hat{\Theta}$ to minimize $\bar{g}_n(\Theta)' W \bar{g}_n(\Theta)$, iterating the weighting matrix W to convergence, and set $\hat{\beta}^\mu = \beta^\mu(\hat{\Theta})$.

6.3 Supply Side

Given $\hat{\Theta}$, we compute model-implied demand and its price derivative under each policy in each market. Substituting these and the observed price into the retailer's pricing first-order condition gives one linear equation in θ_t^c, which we solve to recover the marginal production cost. When only one policy is observed in a market, inversion directly recovers θ_t^c. When both policies coexist, we invert each first-order condition and set θ_t^c to the weighted average of the two implied costs that minimizes the sum of squared pricing residuals.[20] This yields 309 implied cost observations.[21] Regressing θ_t^c on X_t and time fixed effects without an intercept recovers cost parameters $\hat{\alpha}^c$ and store-level cost shocks $\hat{\zeta}_t^c$ from $\theta_t^c = X_t \alpha^c + \zeta_t^c$.

6.4 Parameter Estimates

Table 2 reports selected estimates of $\hat{\Theta}$ and $\hat{\alpha}^c$, with implied $\omega_j = C_j' \hat{\beta}^\omega$ and $\sigma_t = X_t^\sigma \hat{\beta}^\sigma$. Standard errors and 95% percentile confidence intervals are computed from $B = 500$ transaction-level bootstrap replicates. Each draw resamples transactions with replacement and re-estimates the model, recentering GMM moment conditions at their full-sample values evaluated at $\hat{\Theta}$. Bootstrap distributions of the main demand and cost parameters appear in Online Appendix E.1.

Ex-ante valuation dispersion ω_j is widest for dresses (59.217) and narrowest for tops (8.553). The precision intercepts rank in reverse: σ_t is lowest for dresses (0.003) and highest for tops (0.129), indicating that dresses carry the highest ex-post match value uncertainty. Mean utility rises with newly published

within that season, so a larger season raises the denominator and the common seasonal scale does not enter purchase probabilities or ξ_{jt}.

[19] Squaring each market's return- and insurance-rate errors prevents large positive and negative errors from canceling across markets. At the true parameter vector Θ_0 under correct specification, each $m_t(\Theta_0)$ equals zero.

[20] The weights are $(dD_t^r/dp_t)^2$ and $(dD_t^c/dp_t)^2$, because each first-order condition is linear in θ_t^c with slope dD_t^l/dp_t.

[21] All 313 category-month markets enter demand estimation; supply inversion drops four cells where $|\partial D^l/\partial p| < 10^{-5}$, where dividing by a near-zero derivative is numerically unstable.

share (148.054), where a one-percentage-point increase raises μ_t by about 1.5 RMB, reflecting strong consumer preferences for new seasonal releases. Assortment size has little effect on mean utility (−0.125), indicating that additional listings mainly substitute for existing ones rather than drawing new demand into the category. On the supply side, in-season listings have higher marginal production costs (13.429), whereas newly published share has little cost impact (0.099).

Table 2: Selected parameter estimates at $\hat{\Theta}$ and $\hat{\alpha}^c$.

Parameter	Estimate	SE	95% CI
ω_j (dress)	59.217	50.610	[58.965 , 250.506]
ω_j (two-piece)	35.730	6.202	[25.911 , 47.897]
ω_j (outerwear)	35.679	15.850	[19.197 , 83.563]
ω_j (pants)	31.732	23.857	[28.211 , 140.728]
ω_j (sweaters)	21.710	10.912	[16.534 , 74.759]
ω_j (tops)	8.553	4.337	[0.099 , 17.665]
σ_t: newly published share	0.076	0.010	[0.033 , 0.084]
σ_t: dress	0.003	0.000	[0.002 , 0.003]
σ_t: two-piece	0.061	0.009	[0.046 , 0.080]
σ_t: outerwear	0.048	0.010	[0.031 , 0.072]
σ_t: pants	0.050	0.007	[0.020 , 0.056]
σ_t: sweaters	0.052	0.009	[0.017 , 0.068]
σ_t: tops	0.129	0.043	[0.085 , 0.269]
μ_t: Assortment size	-0.125	0.038	[-0.213 , -0.070]
μ_t: Newly published share	148.054	36.079	[146.660 , 327.353]
μ_t: In-season share	2.346	10.220	[-46.314 , 8.125]
θ_t^c: Assortment size	-0.097	0.016	[-0.135 , -0.069]
θ_t^c: Newly published share	0.099	15.580	[-64.999 , 2.317]
θ_t^c: In-season share	13.429	2.184	[12.230 , 22.398]

6.5 Model Fit

Table 3 compares model-implied and sample values of prices, purchase probabilities, and return and insurance rates at the category × month level. Panel A reports unconditional rates per potential consumer, counting both purchasers and non-purchasers in the denominator. These are the moments targeted in estimation,[22] while Panel B reports the corresponding rates per purchase. Each entry is an unweighted mean or standard deviation across the 313 category × month markets.

The model reproduces the key empirical patterns. Mean RI list prices, purchase probabilities, and return rates exceed their CI counterparts in both model and sample, and simulated price levels align with the sample averages.[23] Purchase probabilities are underpredicted by about seven percentage points under both policies.

[22] Estimation targets unconditional rates per potential consumer, which prevents markets with low purchase rates from destabilizing the moment conditions by dividing by near-zero denominators.

[23] In Table 1, pooled raw prices weight each transaction equally. RI transactions place more weight on lower-priced categories such as tops (31% of listings) and less on higher-priced categories such as outerwear than CI transactions do, so pooled CI prices exceed pooled RI prices even though RI prices are higher within every category except tops. Here, each category–month market contributes equally to the mean, downweighting the high-volume categories that drive the pooled ranking.

Table 3: Model vs. sample at $\hat{\Theta}$ (category × month cell means and SDs).

		Mean		SD	
		Model	Sample	Model	Sample
Price	RI (RMB)	107.42	99.89	55.02	39.68
	CI (RMB)	98.33	95.37	49.43	33.41
Purchase	RI (%)	18.99	26.20	20.45	15.51
	CI (%)	15.59	23.04	18.39	12.83
Panel A: Per potential consumer (targeted moments)					
Return	RI return (%)	6.53	7.69	9.50	6.27
	CI return (%)	2.79	5.55	8.41	4.65
	CI insurance (%)	1.39	0.55	8.74	0.79
Panel B: Per purchase (conditional)					
Return	RI return (%)	37.51	29.33	16.88	15.29
	CI return (%)	16.54	22.90	15.74	12.97
	CI insurance (%)	7.26	2.61	20.18	7.77

This systematic underprediction can be due to the true valuation distribution having a heavier tail than the assumed normal.

At $\hat{\Theta}$, unconditional return rates average 6.5% under RI and 2.8% under CI in the model, compared with 7.7% and 5.5% in the sample. In Panel B, conditional return rates miss in opposite directions: the model overpredicts RI returns and underpredicts CI returns. The implied RI-CI gap is 3.7 percentage points in the model and 2.1 percentage points in the sample, compared with 21 and 6.4 percentage points for the conditional gaps in Panel B. Return fit is therefore less poor than the conditional comparison alone suggests, although CI unconditional returns remain underpredicted. Unconditional CI insurance rates are likewise closer than conditional take-up (1.4% vs. 0.6% in Panel A, against 7.3% vs. 2.6% in Panel B). Level fit remains imperfect. We still use the estimates to compare relative welfare across policies within each category, because welfare hinges on the unconditional rates per potential consumer. The error in the unconditional purchase rate is large, but about the same under both policies, so it does not favor either one. The model overstates the unconditional return gap by 1.6 percentage points, which slightly favors CI on the ex-post margin.

7 Counterfactual Simulations

We apply the welfare framework at the estimated parameters to evaluate return policies across categories. We first evaluate a second-best benchmark with efficient returns to isolate the baseline market-structure distortion from monopoly pricing. We then compare the two observed policies, RI and CI, and evaluate an alternative return-penalty discount (RPD) mechanism. Section 7.5 examines the distribution of surplus across agents, and Section 7.6 conducts robustness checks.

7.1 Simulation Design

For the welfare counterfactuals, we construct one *representative* snapshot per category. Covariates, premium-related inputs, demand shocks, and production-cost shocks are averaged across periods using consumer-count weights, and total market size is the sum of those counts. We then solve for equilibrium prices under each policy and evaluate purchase probabilities, return rates, and welfare by aggregating over valuations and coverage. Averaging primitives first, rather than averaging period-by-period welfare, targets long-run performance and keeps noisy thin months from dominating a highly nonlinear welfare functional.

7.2 Second-Best Benchmark

In our model, marginal cost pricing ($p = \theta^c$) and a return refund equal to net salvage recovery $-h$ = scrap value − gross handling implement the first best. On return, the buyer receives net salvage recovery rather than recovering full product price p, fully internalizing the seller's net loss on return in addition to private hassle costs $\kappa + k_i$.[24] Under monopoly, return policy can improve return discipline and participation, but once returns are efficient the remaining distortion is monopoly pricing above θ^c, which stems from market structure.

To isolate this residual market-structure distortion (a loss that return policy design itself cannot correct), we evaluate an *optimal-refund benchmark* (Online Appendix C.2). Setting the consumer refund equal to net salvage value eliminates ex-post return distortions, as return probabilities coincide with the first-best rule q_i^*. The seller chooses p to maximize $(p - \theta^c)D(p)$, because each sale earns $p - \theta^c$ whether the unit is kept or returned. Among single-price policies with a uniform refund, the resulting allocation is second best: the remaining gap to the first best is entirely ex-ante distortion from monopoly pricing. Neither RI nor CI attains this benchmark, because both refund p on return.

This benchmark is a welfare diagnostic rather than a direct policy prescription. Net handling cost h combines logistics, inspection, and resale value that vary across sellers and categories and are imperfectly observed, so mandating refund $-h$ listing by listing is impractical. Furthermore, distance-selling laws generally mandate full product-price reimbursement on no-reason returns, as under China's seven-day no-reason return rules and the EU Consumer Rights Directive.[25] Product-refund forfeiture beyond these statutory floors raises consumer-protection concerns, although restocking fees and partial refunds are tolerated when

[24] Expected seller profit from consumer i is then zero on every path, so $S_i^* = CS_i^*$ and efficient purchase and return decisions maximize consumer surplus as well as social surplus.

[25] China's *Law on Protection of Consumer Rights and Interests* (Art. 25) and *Interim Measures for Seven-Day No-Reason Returns on Online Purchases*. EU Directive 2011/83/EU, Art. 13. Exceptions and prior buyer confirmation narrow these rights.

disclosed at checkout and tied to processing costs.

Table 4 reports the second-best benchmark across the six apparel categories. Efficient returns set $\Delta^{ex\text{-}post} = 0$ by construction, so the remaining welfare shortfall is entirely ex-ante: about 29.4% of first-best welfare for dresses and 23.4%–25.3% in the other five categories, with second-best welfare attaining 70.6% to 76.6% of first best.

Table 4: Second-best benchmark: welfare decomposition and extensive-margin cutoffs

Category	Welfare (% FB)	$\Delta^{ex\text{-}ante}$ (% FB)	Purch. cutoff	Eff. cutoff	Marg. SS
dress	70.60	29.40	56.65	16.13	21.47
outerwear	75.02	24.98	113.81	86.24	26.25
pants	75.13	24.87	91.98	65.55	24.38
sweater	76.59	23.41	110.64	88.93	20.85
top	75.84	24.16	48.61	41.36	7.24
twopiece	74.75	25.25	102.04	76.87	24.21

Columns 4–6 report the purchase cutoff, the first-best efficient cutoff, and marginal social surplus at the purchase cutoff. Marginal social surplus is strictly positive in every category, ranging from 7.24 RMB for tops to 26.25 RMB for outerwear. This confirms that the ex-ante shortfall is monopoly underprovision on the extensive margin: the seller's market power pushes the purchase cutoff above the efficient cutoff, excluding marginal buyers whose purchase would generate positive social surplus. The purchase-efficient gap is largest for dresses (about 40.5) and smallest for tops (about 7.2).

7.3 RI vs. CI

7.3.1 Equilibrium primitives

Table 5 reports equilibrium primitives for RI and CI by category. Columns include prices, production costs, calibrated net return-handling costs, ex-post shock dispersion from σ_t, and model-implied purchase and return probabilities.

Table 5: Category-level equilibrium primitives

						Purchase probability		Return probability	
Category	Production cost	Return handling	Ex-post SD	RI price	CI price	RI	CI	RI	CI
dress	42.06	-48.90	28.95	71.67	68.19	0.11	0.07	0.54	0.34
outerwear	87.73	-76.51	15.77	121.42	115.55	0.27	0.24	0.32	0.11
pants	67.96	-59.55	17.08	100.34	94.27	0.33	0.28	0.37	0.13
sweater	89.99	-72.56	16.64	121.85	113.35	0.43	0.40	0.41	0.15
top	41.39	-25.48	7.58	55.48	49.05	0.17	0.29	0.28	0.04
twopiece	77.95	-65.99	14.65	109.64	103.55	0.21	0.19	0.31	0.10

Notes: Return probabilities are conditional on purchase.

Simulated purchase probability is higher under RI than under CI in every category except tops. This pattern is consistent with the estimation sample, where purchase probability is higher under RI than under CI

at the category-month level on average. Across all categories, the return probability conditional on purchase is lower under CI than under RI.

Category-level variation begins with margins, which dictate the social surplus generated when a product is kept. The dress category exhibits the highest margin, $(p-\theta^c)/p$, at approximately 38%–41% across pricing policies, whereas other categories hover between 25% and 32% under RI (falling to 16% for tops under CI). Economically, a higher margin reflects a wider wedge between ex-ante consumer valuations and production costs. This wedge represents the ex-ante consumption surplus realized upon a kept purchase, implying that potential social gains from expanding market participation are exceptionally high in high-margin segments.

When a consumer returns the product, the welfare outcome depends on the net return-handling cost (h_t), which captures salvage value net of shipping and processing frictions. Although a high return probability typically generates ex-post distortions, this effect is moderated by the efficiency of the return channel. In the dress category, net handling cost is $h_t = -48.90$ RMB (a net salvage recovery of 48.90 RMB), preserving 68%–72% of the retail price, so the social cost of a return is relatively low. For tops, where margins are lower, net handling cost is −25.48 RMB (recovering 25.48 RMB), which covers only 46%–52% of the price, so the social cost of a return is much higher.

Ultimately, the return probability weighs these two competing forces: a higher return rate shifts the welfare weight from ex-ante consumption surplus to ex-post salvage recovery, with the net outcome depending on whether the margin and net salvage recovery dominate the welfare-reducing frictions of returns.

7.3.2 Welfare decomposition

Figure 5 shows how these primitives aggregate into welfare. Each stacked bar expresses welfare and the shortfalls from ex-post and ex-ante distortions as percentages of first-best welfare. Ex-post distortion is larger under RI than under CI in every category, consistent with the lower return probabilities under optional insurance. Full-forfeiture RPD is included for later comparison.

On the ex-post margin, both RSI schemes refund p on return and are more lenient than the optimal-refund benchmark. On return, the seller incurs net cost $p + h$ (refunding p and absorbing net handling cost h), and the insurer bears return shipping k_i when reimbursed. The uninternalized loss imposed on the seller and the insurer upon return is therefore $p + h + k_i$ under RI and for insured CI purchasers, but only $p + h$ for uninsured purchasers who bear k_i themselves. Holding price fixed, RI therefore imposes the larger gap on every purchaser, whereas under CI only insured buyers face $p + h + k_i$. Because price is endogenous, it

can be higher under CI. Provided the equilibrium price under CI does not exceed that under RI by too much and insurance take-up is not too large, CI generally achieves better ex-post return discipline than RI. Our simulation falls in this case.

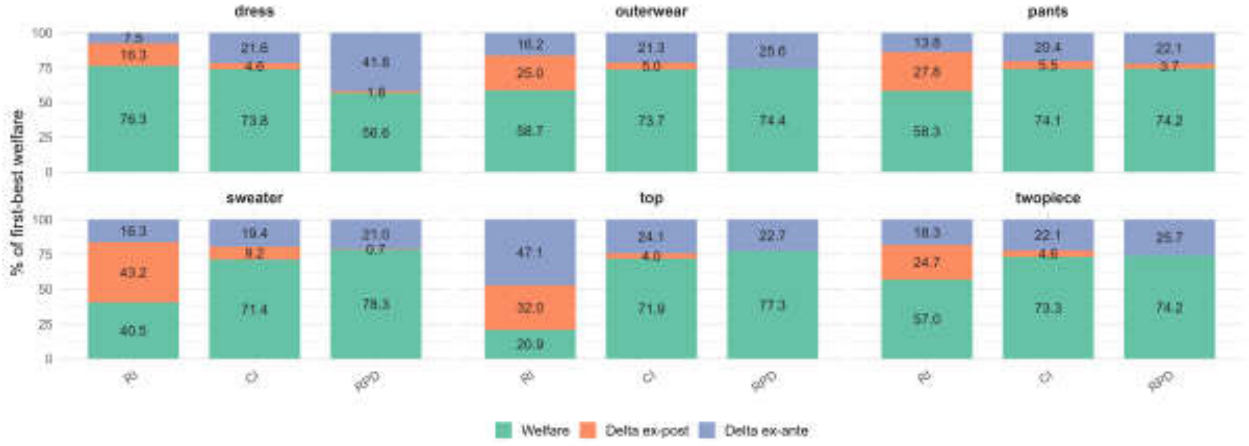


Figure 5: Welfare decomposition by category, normalized as a share of first-best welfare.

For dresses, RI attains a larger overall share of first-best welfare than CI (76% vs. 74%). Automatic coverage attracts additional buyers who remain socially valuable on net, so the gain in ex-ante efficiency dominates the incremental ex-post distortion from returns.[26] Outside the dress category, the ranking reverses. When margins are lower and return costs account for a larger share of surplus per kept sale, the gain from curbing ex-post over-return dominates the gain from additional participation, so CI achieves higher welfare. RI is especially harmful in thin-margin categories such as tops, where CI outperforms it on both the ex-ante and ex-post margins. Roughly 80% of the first-best benchmark for tops is dissipated under RI in the decomposition, driven by poor ex-ante screening and ex-post over-return.[27]

7.3.3 Extensive versus ex-post margin trade-off

On the extensive margin, RSI expands participation relative to the optimal-refund benchmark and CI. Participation expands for two reasons. A full product refund together with return-shipping reimbursement raises willingness to buy, and higher return rates under RI weaken the surplus-extraction pricing incentive.

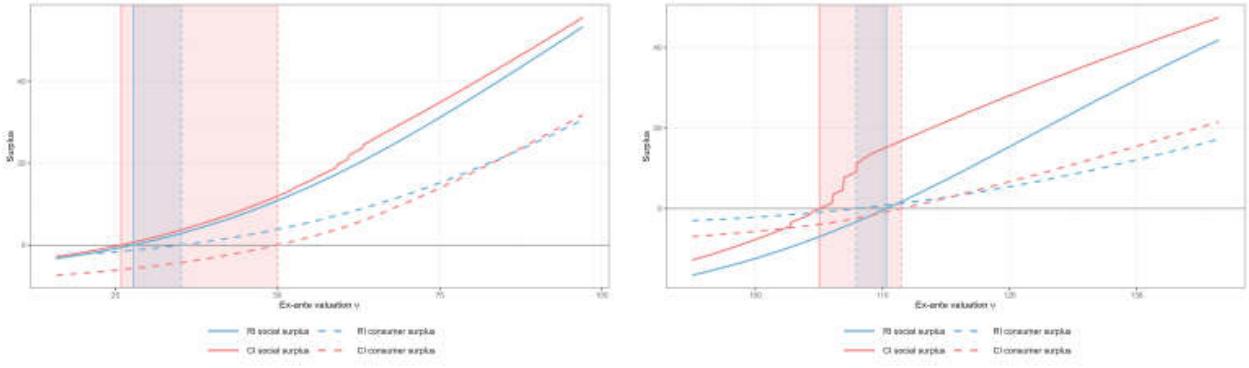
The seller's pricing pushes the purchase cutoff above the efficient cutoff, excluding marginal buyers with positive social surplus. To quantify this extensive-margin underprovision, we use the purchase-efficient gap defined in Section 4.3. In a textbook single-product monopoly without returns, the degree of underprovision

[26] Ex-post uncertainty need not imply a higher marginal *social* value of additional participation: greater dispersion in ex-post valuations raises private option value but also tends to increase return probabilities, and once return-handling and refund costs are accounted for, the social surplus contributed by marginal entrants need not rise mechanically with uncertainty.

[27] The model overstates the RI–CI return gap relative to the sample (Table 3), so CI's estimated advantage outside dresses is, if anything, overstated, while RI's advantage in dresses is conservative.

increases with the margin and the price. Across return schemes, however, posted price and margin poorly measure extensive-margin distortions. RI and CI attach different return benefits and insurance costs for buyers and shift costs differently for sellers, so they are effectively different products rather than the same good. The gap is therefore a better metric for comparing extensive-margin underprovision across policies.

Figure 6 illustrates the extensive-margin comparison for dresses and outerwear at estimated equilibria. Solid curves are individual social surplus and dashed curves are consumer surplus, each evaluated at the equilibrium price. Solid vertical lines mark efficient cutoffs where social surplus is zero, and dashed vertical lines mark purchase cutoffs where consumer surplus is zero. The purchase-efficient gap (the shaded band) measures extensive-margin distortion. Curves and cutoffs average across empirical coverage masses.



(a) Dress. (b) Outerwear.

Figure 6: Extensive-margin cutoffs under RI and CI. Under CI, social surplus jumps at insurance cutoffs because coverage lowers private return cost. With multiple coverage levels, the averaged curve can appear kinked.

Relative to CI, whether RI performs better depends on whether extensive-margin gains from broader participation outweigh ex-post losses from over-return. For dresses, both policies underprovide, but CI's gap is much larger (about 24 versus 8 under RI). In Figure 6, the vertical difference between the solid curves of CI and RI captures the combined effect of equilibrium pricing and the reduction in ex-post return distortions under CI when buyers remain uninsured. For dresses, CI's solid curve lies only modestly above RI's, so the ex-post margin gain from having uninsured buyers bear return shipping is small. This modest difference between the two solid curves reflects the large ex-post shock dispersion implied by σ_t (an ex-post standard deviation of 28.95 RMB in Table 5). When match value uncertainty is high, idiosyncratic taste shocks dominate the deterministic return shipping fee k_i, dampening the behavioral response of returns to being uninsured. Against that small ex-post gain, RI's smaller extensive-margin gap dominates. The wide band of socially valuable buyers between the two purchase cutoffs raises welfare relative to CI. The purchase cutoffs differ substantially because high return rates make shipping coverage especially valuable for dresses. RI

provides this coverage automatically, whereas CI requires consumers to pay a separate premium.

For outerwear, the extensive-margin distortion differs between RI and CI. CI still underprovides, whereas RI overprovides. The margin trade-off reverses relative to dresses. Across consumers' valuations, CI's solid curve lies well above RI's because the ex-post shock dispersion implied by σ_t is much smaller. When match value uncertainty is small, having uninsured buyers bear return shipping costs strongly disciplines their return behavior and hence greatly improves ex-post efficiency. On the other hand, RI's broader participation adds little on the extensive margin. The RI-CI difference in purchase cutoffs is narrow, and some of the buyers RI admits lie in the overprovision band. Return discipline under CI therefore dominates. Under RI, overprovision can arise because the seller does not fully internalize insurer claim losses, admitting marginal, return-prone consumers whose transactions generate negative social surplus.

Table 6 reports the same cutoffs and marginal social surplus for all categories and adds RPD next to RI and CI, matching the objects in Table 4. The remaining categories follow outerwear: CI underprovides and RI overprovides. Relative to the second-best gaps in Table 4 (about 7 to 40), RSI softens underprovision on the extensive margin, and RI softens more than CI and can overshoot. The RI gap falls to about 8 for dresses and turns negative outside dresses (about -1.5 to -6), while CI gaps stay positive outside dresses (about 4 to 6).

Table 6: Purchase cutoffs, efficient cutoffs, and marginal social surplus by category

	Purchase cutoff			Efficient cutoff			Marginal social surplus		
Category	RI	CI	RPD	RI	CI	RPD	RI	CI	RPD
dress	35.19	50.02	66.71	27.45	25.56	35.08	2.87	11.96	26.48
outerwear	107.94	111.44	114.17	110.28	105.70	87.71	-3.27	16.76	26.45
pants	84.87	89.28	90.30	87.31	83.02	76.97	-2.93	15.48	16.19
sweater	107.07	108.66	109.49	113.17	104.26	94.94	-8.87	11.02	15.76
top	52.16	48.60	48.36	53.64	43.91	41.38	-3.33	5.93	6.98
twopiece	97.80	100.14	102.32	100.01	94.23	77.93	-3.32	15.94	24.37

Notes: Entries average across coverage types. Efficient cutoffs use each scheme's equilibrium return behavior.

The CI gap exceeds RI's for two reasons. First, when marginal entrants remain uninsured, stronger ex-post discipline raises their social surplus relative to RI and can lower the efficient cutoff. Second, the purchase cutoff itself is higher than under RI. Both enlarge the purchase-efficient gap. CI reduces return distortions but can leave more socially valuable purchases unrealized.[28]

Dresses and the moderate-margin categories (outerwear, pants, sweater, and two-piece) illustrate the trade-off between extensive-margin participation and ex-post return discipline. Thin-margin categories

[28] When marginal entrants would adopt CI coverage instead, ex-post behavior coincides with RI and the efficient cutoff is unchanged. The higher purchase cutoff nevertheless still enlarges the gap.

such as tops demonstrate that the purchase and return margins can move in the same destructive direction. RI destroys surplus across both margins through simultaneous ex-post over-return and extensive-margin overprovision.[29]

7.4 Return-Penalty Discount (RPD)

Under return-penalty discount (RPD), each consumer chooses either a discounted price with no or partial product refund on return, or a full price with full refund rights, as in common "final sale" or restocking-fee menus. Amazon allows a restocking fee of up to 20 percent after the return window. Taobao and JD.com implement China's seven-day no-reason return as a full refund of the product price. RPD preserves a full-refund option while adding a second branch with stricter refund terms. On the discount branch, share $\phi \in (0, 1]$ of the discounted price is forfeited on return. Equilibrium conditions and the explicit expressions for the welfare objects appear in Online Appendix C.1.

Relative to RSI, RPD is an alternative for correcting excessive returns, especially when margins are thin and social return costs are high. RSI expands low-end participation by subsidizing returns, thereby creating adverse selection ex-ante and over-return ex-post. In contrast, RPD makes the discount attractive to high-valuation keepers rather than to buyers who expect to return, so it does not expand low-end participation through return subsidies. On that branch, forfeiture also lowers the return probability ex-post.

RPD also places screening with the party that holds product-level information. Under CI, the insurer sets standardized premiums while the seller chooses only product price. Even when the return rate is low, a premium set from return probability can be too low if handling costs are high, so optional coverage underprices the social cost of returns. Cheap coverage can then generate additional purchases and returns that impose a large negative externality on the seller, which cannot be corrected through product price alone. In contrast, under RPD the platform sets the refund terms, and the seller chooses both the list price and the discount. These two instruments let the seller incorporate private information about return costs into the menu, which can improve welfare relative to CI. Nevertheless, the additional instruments can also strengthen the seller's ability to extract surplus and enlarge the purchase-efficient gap.

[29] These rankings use each scheme's private return behavior q_i^l. The second-best efficient cutoff under q_i^* remains below every q_i^l-efficient cutoff, so the first-best residual $\Delta^{ex\text{-}ante}$ is still underprovision even where RI overprovides on the extensive margin for some categories (Table 4).

7.4.1 Full-forfeiture RPD

We first report full-forfeiture RPD ($\phi = 1$), the leading case that delivers the strictest return discipline and is straightforward to implement. Ex ante, stricter refund terms can exclude socially valuable marginal buyers. Ex post, no product refund narrows the return subsidy to an externality gap of $-h$ alone. When scrap recovery exceeds handling ($h < 0$), this gap is positive, so full forfeiture can overshoot into under-return because consumers do not internalize the positive externality that net recovery confers on the seller. Table 7 reports the list price, discount d, effective price, and segment-level purchases and returns.

Depending on equilibrium prices and the wedge between the purchase cutoff and the discount-to-full-price segmentation cutoff, buyers may end up only on the discount branch. In most categories full-price demand is negligible, purchases concentrate on the *discount* branch, and return probability there is close to zero. In contrast to RSI, ex-post distortion takes the form of under-return rather than over-return. Ex-post distortion is nevertheless smaller under full-forfeiture RPD than under either RSI policy (Figure 5).

Table 7: RPD counterfactual: prices and segment-level purchase probabilities, returns, and return probabilities by category

				Discount option			Full price		
Category	Price	Discount	Effective price	Purchase prob.	Return	Return prob.	Purchase prob.	Return	Return prob.
dress	105.00	37.70	67.30	0.04	0.00	0.01	0.00	0.00	
outerwear	202.10	87.93	114.17	0.22	0.00	0.00	0.00	0.00	
pants	95.30	2.40	93.79	0.17	0.00	0.00	0.10	0.02	0.24
sweater	114.56	4.90	109.77	0.38	0.00	0.00	0.01	0.00	0.34
top	49.06	0.69	48.36	0.30	0.00	0.00	0.00	0.00	
twopiece	181.15	78.83	102.32	0.18	0.00	0.00	0.00	0.00	

Notes: Effective price is $p - d$. Full-price return probabilities are left blank for segments with negligible full-price volume.

Participation, on the other hand, is weaker. By driving return probabilities on the discount branch close to zero, refund forfeiture leaves the seller with a pool of confident keepers who rarely return, sharpening the surplus-extraction incentive. For dresses, the RPD purchase cutoff is about 67, against about 50 under CI and 35 under RI, even though RPD's effective price (about 67) lies slightly below the CI price (about 68). Forfeiting the product payment on return removes the implicit full-refund subsidy, so the purchase cutoff can rise above the CI cutoff even when the demand-weighted price paid is lower.

Relative to the RI and CI purchase-efficient gaps above, full-forfeiture RPD underprovides by more. For dresses the gap rises to about 32 and marginal social surplus to about 26 (Table 6). Outside dresses the same ordering generally holds. RI often overprovides, CI underprovides, and RPD underprovides by more.

This stronger screening still raises welfare relative to RI and CI in every category except dresses. In the dress category the extensive margin is especially valuable. Marginal entrants often have relatively low valuations and high return propensity, yet many such purchases remain socially worth encouraging because

kept sales generate high surplus. RSI preserves a channel for those buyers. The discount branch under full-forfeiture RPD largely shuts it down, so the participation loss outweighs the ex-post gain.

7.4.2 Partial forfeiture and restocking fees

Full forfeiture is extreme. Softening the return penalty on the discount branch to a partial forfeiture $\phi \in (0, 1)$ restores some participation while retaining most of the return discipline. A natural retail implementation is a restocking fee. On return the buyer pays $\phi(p - d)$ and recovers share $1 - \phi$ of the discounted payment. We focus on restocking fees in the 5–20% range, which are common in retail practice on eligible returns. Figure 7 shows the welfare decomposition across categories as ϕ varies over this range.

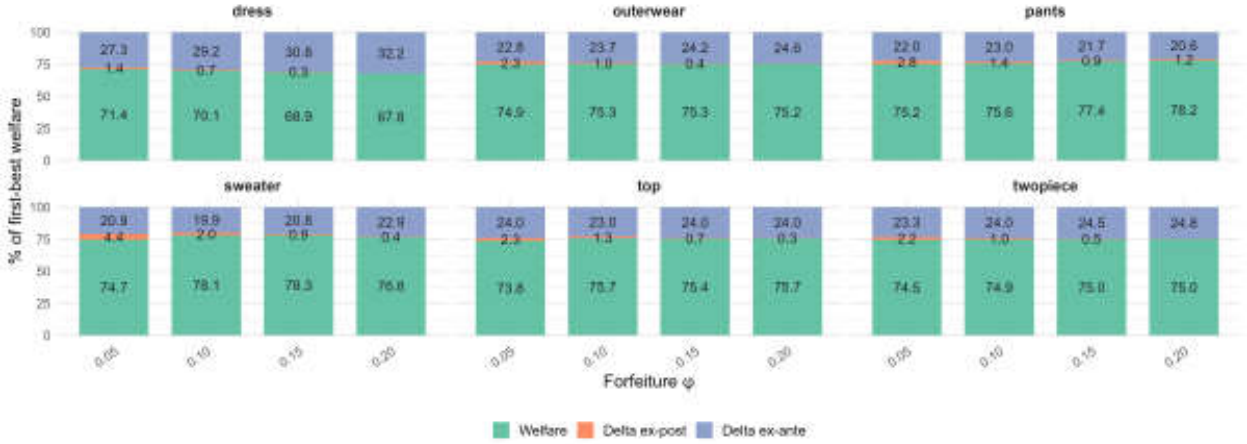


Figure 7: Partial-forfeiture RPD: welfare decomposition by category and $\phi \in \{0.05, 0.10, 0.15, 0.20\}$, normalized within each cell.

Lower ϕ reduces under-return and raises participation relative to full forfeiture while retaining most of the return discipline from refund forfeiture on that branch. At $\phi = 0.10$–0.15, welfare exceeds both RSI policies in every category other than dresses and matches or exceeds the optimal-refund benchmark, attaining about 75–78% of first-best compared with roughly 75–77% under that benchmark. Welfare can exceed the second-best benchmark because RPD provides a two-option screening menu (second-degree price discrimination across return propensities), whereas the second-best benchmark is restricted to a single posted price. For dresses, welfare remains below RI (about 76% of first-best) but at low ϕ approaches the optimal-refund benchmark (about 71% at $\phi = 0.05$–0.10) and lies well above full-forfeiture RPD (about 57%). A low fee (about 5–10% of $p - d$) limits the participation loss from full forfeiture, but RI still ranks first. For tops, where margins are thin and curbing ex-post return distortion and inefficient purchases on the extensive margin matters most, full-forfeiture RPD dominates both RSI policies and partial-forfeiture RPD in the 5–20% band. Elsewhere, welfare peaks in that band, so moderate partial-forfeiture RPD outperforms

both RSI policies and full-forfeiture RPD. The pattern favors category-specific fees over a single platform rule.

7.5 Welfare across agents

The decompositions above split efficiency losses relative to first best. Figure 8 instead shows how expected social surplus divides among consumers, the seller, and the insurer, as shares of first-best welfare. Partial-forfeiture RPD uses the welfare-maximizing restocking fee. Expected consumer surplus and seller profit lie above zero. Insurer profit is at or below zero. The three components sum to welfare.

Within each category, the ranking of policies by welfare, expected consumer surplus, and seller profit is broadly the same. Insurer profit is zero under RPD, which has no insurance, and under CI outside dresses, where no one buys coverage. When buyers are insured, expected claims exceed the observed premiums, so insurer profit is negative under RI in every category and under CI only for dresses.[30]

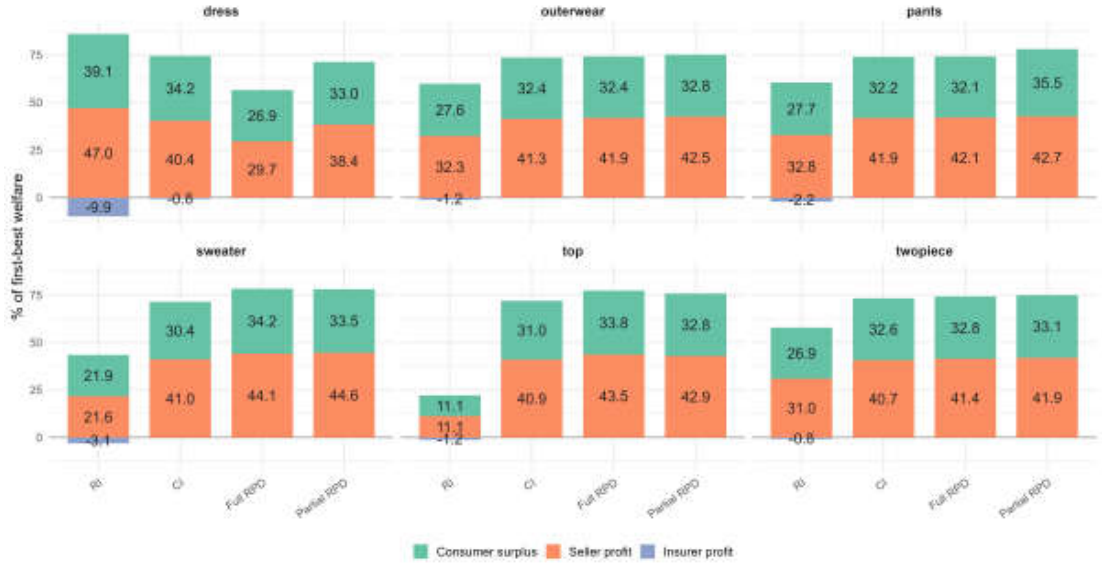


Figure 8: Welfare across agents by category and return policy.

RPD and CI deliver similar average expected consumer surplus, but they allocate it differently. RPD's stricter refund raises the purchase cutoff and excludes many low-valuation buyers, who then receive zero surplus. Under CI the product price is usually lower, and those buyers can still purchase optional coverage, which can raise their expected consumer surplus relative to RPD.

[30]Negative insurer profit for a single apparel seller is consistent with reported underwriting losses on Taobao return-shipping insurance (Huatai in 2012, Wang Liangdi, 2017, Peking University; ZhongAn in 2014–2017, Tu Yinghao, *National Business Daily*, 2020). Insurers can still break even by spreading risk across merchants, cross-subsidizing high-return apparel with low-return categories, or offsetting category losses through their relationship with the platform, so long as aggregate profit across merchants is non-negative.

7.6 Robustness check

We next assess sensitivity to the scrap-value calibration and sampling uncertainty in welfare rankings and decompositions.

7.6.1 Scrap-value sensitivity

The baseline decomposition uses the scrap calibration in Section 6.1 (about 75% of price on average). Because $h_t = 15 -$ scrap value, alternative recovery rates shift the return externality and can reorder counterfactuals even when demand primitives are fixed. Online Appendix D recomputes welfare at scrap shares of $0.65\,p_t$ and $0.85\,p_t$.

When recovery is low (65%), returns are costlier, ex-post distortion under RI rises, and CI ranks above RI in every category, including dresses. The baseline RI advantage for dresses therefore does not survive. When recovery is high (85%), RI still beats CI only for dresses, while the ranking between CI and full-forfeiture RPD splits by category. CI is ahead for dresses, outerwear, pants, and two-piece. Full-forfeiture RPD is ahead for sweaters and tops. Full-forfeiture rankings are therefore not invariant to scrap-recovery assumptions. The qualitative lesson is nevertheless stable. Reimbursement-heavy policies fare worse when recovery is weak and better when recovery is strong.

At 65% and 85% scrap, partial-forfeiture RPD largely confirms the baseline pattern. Outside the dress category, $\phi \approx 0.05$–0.20 still beats both RSI policies at both scrap shares. For dresses, partial-forfeiture RPD at 85% scrap remains below both RSI policies even at low ϕ, and RI ranks first, as at baseline. At 65% scrap it beats RI but remains close to CI.

7.6.2 Inference on welfare ranking and decomposition

Online Appendix E reports bootstrap inference for welfare levels and decompositions by policy within each category. For most objects, the bootstrap histograms are approximately symmetric and close to normal around the full-sample estimate, which is expected under the recentered bootstrap. Table 8 in Online Appendix E reports pairwise bootstrap win rates for actual welfare.

Welfare rankings are largely robust across categories. RI continues to rank above the other policies for dresses, while partial-forfeiture RPD ranks highest outside the dress category when implemented at the welfare-maximizing restocking fee over $\phi \in \{0.05, 0.10, 0.15, 0.20\}$. In some categories, such as two-piece and outerwear, the welfare distributions under CI, full-forfeiture RPD, and partial-forfeiture RPD are difficult to distinguish by eye. Visual comparisons should therefore not be over-interpreted. The decomposition terms

are nevertheless informative and align with the mechanisms discussed above. Relative to CI, RI exhibits lower ex-ante distortion but higher ex-post distortion in all categories except tops, where it performs worse on both margins. Partial-forfeiture RPD substantially reduces ex-ante distortion relative to full forfeiture while delivering similar ex-post distortion, or only slightly higher.

Overall, the bootstrap evidence corroborates the welfare rankings and decomposition patterns discussed above.

8 Conclusion

This paper studies the welfare consequences of RSI and RPD in online retail. Using transaction-level data from an apparel retailer on Taobao that repeatedly switched between RI and CI, we estimate a structural model of purchase, return, and insurance decisions under monopoly pricing across six garment categories. We show that the welfare ranking among return policies is category-specific, governed by a trade-off. When the profit margin is moderate to high, a generous return policy improves ex-ante efficiency at the cost of ex-post efficiency. By contrast, in thin-margin categories such as tops, a stricter return policy improves ex-ante and ex-post efficiency simultaneously.

The optimal policy depends on margins and return-handling costs. For dresses, high margins and high salvage recovery imply that the surplus from the additional efficient purchases under RI outweighs the cost of the additional returns, so RI achieves the highest social surplus. In thinner-margin categories such as tops, margins are low and salvage recovery is poor, so reverse logistics dissipate the surplus from kept sales. CI or RPD then outperforms RI by screening out inefficient purchases and tightening return discipline. When platforms or retailers cannot tailor return rules to category-specific primitives, partial-forfeiture RPD with moderate restocking fees (10%–15%) provides a practical, robust alternative that balances market participation against return control.

These findings challenge prevailing e-commerce platform practices. Platforms frequently compel merchants to offer automatic return insurance indiscriminately, even on low-cost, thin-margin merchandise, or require them to adopt RI in order to join major sales events such as Singles' Day. Our results demonstrate that mandatory automatic coverage is inefficient in thin-margin categories, where up to 80% of first-best welfare is dissipated through over-return and unselective entry. The social cost of uniform reimbursement is magnified under emerging platform practices such as refund-without-return policies, where the buyer keeps the merchandise and still receives a refund. The seller then bears the full disposition cost, so handling costs can

exceed our baseline calibration. Rather than imposing a uniform reimbursement mandate, platforms should either allow return policies to reflect category margins and reverse logistics costs or deploy a self-selective RPD menu.

8.1 Limitations and future research

We treat RSI as given when recovering demand and inverting seller pricing. Conditional on the observed scheme, endogenizing enrollment would leave demand estimation largely unchanged. Purchase, insurance, and return still identify the same primitives. The supply side would change, because optimality of the chosen scheme becomes an additional inequality constraint on total profit at the store level. Recovered costs would have to rationalize both within-scheme pricing and the seller's preference for RI versus CI at the store level. We deliberately avoid that inequality.[31] Total profit at the store level depends on the relative market size of each category, whereas production cost recovered only from the observed price does not. The model treats each category on its own, and the welfare comparison is within a category, so relative market size should not enter the recovered costs.

Several extensions would change the policy ranking. First, under oligopoly, buyers screened out by a high price or a strict refund can divert to other products or sellers. That diversion mitigates the social cost of exclusion relative to our single-seller benchmark and therefore tilts welfare toward stricter return terms. Second, we study platform-regulated refund menus (RPD). Delegating refund choice to sellers adds a screening instrument that can expand participation for some types while strengthening surplus extraction, with an ambiguous net welfare effect. Third, our counterfactuals are risk-neutral; risk aversion would raise the value of automatic coverage. Fourth, we take insurer premiums as exogenous. Endogenizing them would reintroduce double marginalization between the insurer and the seller, with an ambiguous effect on the ranking across categories.[32] Fifth, on a two-sided platform, return policy also shifts cross-network externalities between buyers and sellers; characterizing the platform-optimal refund then requires modeling both sides jointly, which we leave for future research.

[31] Both schemes are observed in 18 of 73 months. In those months the store changes scheme while category fundamentals are expected to be relatively unchanged, so a ranking of total profit cannot justify the observed scheme.

[32] Double marginalization pushes the retail price above the vertically integrated benchmark. The wedge is weaker than in a fixed-cost insurance model because expected claims scale with the return probability, so the insurer sets a lower premium than when claims are a fixed cost. Remaining pass-through can still widen the private-social gap in return costs. Ex ante, higher prices are costly when marginal purchases have positive social surplus (as for dresses) but can help in thin-margin categories where RSI already overprovides. Ex post, a higher price can worsen over-return.

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Online Appendix

A Data Construction

This appendix constructs the analysis sample from raw transactions in three steps: filters, RSI classification and CI imputation, and aggregation.

A.1 Sample Filtering

We keep 2017–2023 transactions and drop multi-item or multi-quantity orders (complementarity and lower marginal return cost), orders without a logistic ID (typically cancelled before shipment), orders with buyer-paid premium shipping services (a separate surcharge outside the product price), partial refunds for minor defects, and non-product listings. We also drop single-day RSI switches for promotions such as Singles' Day, so the sample reflects the store's usual scheme.

A.2 RSI Scheme Classification and CI Imputation

Missing insurance fields are ambiguous (RI ineligibility versus CI non-purchase). Complete records with premium and coverage are RI. Coverage only is CI. We propagate the day-level majority label, treat long runs of missing records as CI, and drop remaining ambiguous cases and confirmed RI ineligibility (1.8% of RI transactions).

CI coverage is imputed from the nearest same-province observation by date. Premiums are not recorded in 2017–2023, so we impute $I^c(k_i) = ak_i + b$ from 2025 Taobao data. Agents do not internalize current returns into future premiums.[33]

A.3 Aggregation, Filtering and Smoothing

We aggregate the cleaned transactions to category × month cells, counting sales and averaging transaction prices within each cell. We keep categories whose average monthly sales over the sample reach at least fifty transactions. The six qualifying categories are dress, tops, pants, sweater, two-piece, and outerwear.

Monthly sales and prices are smoothed with a five-month centered moving average that combines the current month with the two adjacent months on each side, using the available months only at the ends of the sample. Each category-month is then resampled to its smoothed sales count, drawing from the month's observed transactions without replacement when observed sales exceed the smoothed count and

[33] We treat the 2025 premium-coverage slope as stable. It implies premiums of roughly 60% of coverage, consistent with older public documentation (e.g. Zhihu).

with replacement otherwise. Every resampled transaction in a category-month is assigned the smoothed price, so RI and CI transactions in the same month share one list price and therefore one net handling cost $h_t = 15 - 0.75\, p_t$.

Market size is twice the peak smoothed monthly sales within the category-season, with seasons defined as March–May, June–August, September–November, and December–February. Monthly purchase probabilities equal smoothed sales divided by market size. We drop category-months with purchase probability below 5% to avoid the zero-share problem.

When both schemes appear within a month, we compute each scheme's time share as the share of the month's transaction-active hours during which the scheme was in effect, common across categories. The observed purchase, return, and insurance-purchase counts under each scheme are divided by that time share, converting them into full-month counts at the common smoothed price. The scale factor is capped at the market size divided by twice the scheme's observed transaction count, so no scheme's full-month volume exceeds half the market size and purchase probabilities remain interior. Because a single factor multiplies every coverage × purchase × return (× insurance) cell of the scheme-month, the scaling preserves the return rate per purchase and the insurance take-up rate within the scheme-month.

B Identification of Ex-Post Precision

This appendix derives the expression for expected returns under RI and proves that, holding the purchase schedule fixed, expected returns are strictly decreasing in ex-post precision σ_t.

Under RI, return shipping is fully reimbursed up to k_i, yielding mean return utility $\overline{u}_{it}^{\text{return}} = -\kappa$. A consumer purchases if and only if $\nu_{it} \geq \hat{\nu}_t^r \equiv p_t + \frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t\kappa})$. Defining willingness to pay as $w_{it}^r \equiv \nu_{it} - \frac{1}{\sigma_t}\ln(1-e^{-\sigma_t\kappa})$, the purchase condition is equivalent to $w_{it}^r \geq p_t$. Because $\nu_{it} \sim \mathcal{N}(\mu_t + \xi_t, \omega_j^2)$, w_{it}^r is normally distributed with mean $\bar{w}_t^r = \mu_t + \xi_t - \frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t\kappa})$ and dispersion ω_j.

Conditional on purchase, the probability of return is

$$q_{it}^r(\nu_{it}) = \frac{e^{-\sigma_t\kappa}}{e^{\sigma_t(\nu_{it}-p_t)} + e^{-\sigma_t\kappa}}.$$

Substituting $\nu_{it} - p_t = (w_{it}^r - p_t) + \frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t\kappa})$ into the denominator and dividing both numerator and denominator by $e^{-\sigma_t\kappa}$ simplifies the individual return probability to

$$q_{it}^r(w_{it}^r) = \frac{1}{1 + (e^{\sigma_t\kappa} - 1)e^{\sigma_t(w_{it}^r - p_t)}}. \tag{15}$$

Integrating over the distribution of w_{it}^r yields the unconditional expected return rate per potential buyer:

$$\mathbb{E}[R_{it}^r(p_t)] = \mathbb{E}_{w_{it}^r}\left[\frac{1}{1+(e^{\sigma_t\kappa}-1)e^{\sigma_t(w_{it}^r-p_t)}}\,\mathbf{1}\big(w_{it}^r \geq p_t\big)\right]. \tag{16}$$

The return rate conditional on purchase is simply $\mathbb{E}[R_{it}^r(p_t)]/\mathbb{E}[D_{it}^r(p_t)]$.

As shown in Section 5, price variation identifies ω_j and $\bar{w}_t^r$, holding the distribution of w_{it}^r and the purchase probability $\mathbb{E}[D_{it}^r(p_t)] = 1 - \Phi\big(\frac{p_t-\bar{w}_t^r}{\omega_j}\big)$ fixed. For every purchaser ($w_{it}^r \geq p_t$), $w_{it}^r - p_t \geq 0$. Because $\kappa > 0$, the term $e^{\sigma_t\kappa} - 1 > 0$ is strictly increasing in σ_t, while $e^{\sigma_t(w_{it}^r-p_t)} \geq 1$ is weakly increasing in σ_t. The denominator of (15) is therefore strictly increasing in σ_t, so each purchaser's return probability is pointwise strictly decreasing in σ_t. Because the distribution of w_{it}^r and the purchase region $w_{it}^r \geq p_t$ are held fixed, the expected return rate (16) (as well as the return rate conditional on purchase) is strictly decreasing in σ_t. Consequently, σ_t is uniquely identified by the return rate, after which intrinsic mean valuation $\mu_t + \xi_t = \bar{w}_t^r + \frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t\kappa})$ is uniquely recovered.

C Return-Policy Extensions

C.1 Return-Penalty Discount: Equilibrium

RPD offers a discount branch at $p - d$ with forfeiture share $\phi \in [0, 1]$ of $p - d$ on return, or a full-price branch at p with a full product refund. The buyer bears $\kappa + k_i$ on either return; the discount branch recovers $(1 - \phi)(p - d)$.

Expected surplus on the two branches is

$$CS_i^f = \frac{1}{\sigma}\ln\left(e^{\sigma(\nu_i - p)} + e^{-\sigma(\kappa+k_i)}\right), \tag{17}$$

$$CS_i^d = d - p + \frac{1}{\sigma}\ln\Big(e^{\sigma\nu_i} + e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}\Big). \tag{18}$$

The consumer prefers the discount branch when $CS_i^d \geq CS_i^f$, which defines

$$\tilde{\nu}^{RPD} = -(\kappa + k_i) + \frac{1}{\sigma}\ln\left(\frac{e^{\sigma(p-d)} - e^{\sigma(1-\phi)(p-d)}}{1 - e^{-\sigma d}}\right). \tag{19}$$

Purchase thresholds are

$$\hat{\nu}^d = \frac{1}{\sigma}\ln\Big(e^{\sigma(p-d)} - e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}\Big), \tag{20}$$

$$\hat{\nu}^f \equiv p + \frac{1}{\sigma}\ln\big(1 - e^{-\sigma(\kappa+k_i)}\big). \tag{21}$$

The consumer purchases when $\max\{CS_i^d, CS_i^f\} \geq 0$. Both branches are active when $\tilde{\nu}^{RPD} \geq \hat{\nu}^f$, or

equivalently (when $1 - e^{-\sigma d} > 0$)

$$\frac{e^{\sigma(p-d)} - e^{\sigma(1-\phi)(p-d)}}{1 - e^{-\sigma d}} \geq e^{\sigma p}\left(e^{\sigma(\kappa+k_i)} - 1\right). \tag{22}$$

Conditional on purchase,

$$q_i^d = \frac{e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}}{e^{\sigma\nu_i} + e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}}, \tag{23}$$

$$q_i^f = \frac{e^{-\sigma(\kappa+k_i)}}{e^{\sigma(\nu_i-p)} + e^{-\sigma(\kappa+k_i)}}. \tag{24}$$

With branch indicators D_i^d, D_i^f and aggregates D^d, R^d, D^f, R^f, the seller chooses (p, d) to maximize $\pi^{RPD} = \pi^d + \pi^f$, where $\pi^d = (p - d - \theta^c)D^d - [(1-\phi)(p-d) + h]R^d$ and $\pi^f = (p - \theta^c)D^f - (p + h)R^f$. Individual social surplus on each branch is

$$S_i^d \equiv CS_i^d + (p - d - \theta^c) - q_i^d\left[(1-\phi)(p-d) + h\right], \tag{25}$$

$$S_i^f \equiv CS_i^f + (1 - q_i^f)(p - \theta^c) - q_i^f(\theta^c + h). \tag{26}$$

Define $CS^{RPD} \equiv \mathbb{E}_{\nu_i,k_i}[D_i^d\, CS_i^d + D_i^f\, CS_i^f]$ and $W^{RPD} \equiv \mathbb{E}_{\nu_i,k_i}[D_i^d\, S_i^d + D_i^f\, S_i^f]$. The main-text full-forfeiture case is $\phi = 1$; partial forfeiture varies ϕ.

C.2 Optimal Refund Policy

With list price p and return refund $-h$,

$$CS_i^{\text{rs}} = -p + \frac{1}{\sigma}\ln\left(e^{\sigma\nu_i} + e^{-\sigma(h+\kappa+k_i)}\right), \tag{27}$$

$$\hat{\nu}^{\text{rs}} = \frac{1}{\sigma}\ln\left(e^{\sigma p} - e^{-\sigma(h+\kappa+k_i)}\right), \tag{28}$$

where $\hat{\nu}^{\text{rs}}$ solves $CS_i^{\text{rs}} = 0$ when $e^{\sigma p} > e^{-\sigma(h+\kappa+k_i)}$. The consumer purchases when $\nu_i \geq \hat{\nu}^{\text{rs}}$ and returns with $q_i^{\text{rs}} = q_i^*$ from (11). Aggregates are

$$D^{\text{rs}} \equiv \mathbb{E}_{\nu_i,k_i}\left[\mathbf{1}\left(\nu_i \geq \hat{\nu}^{\text{rs}}\right)\right], \tag{29}$$

$$R^{\text{rs}} \equiv \mathbb{E}_{\nu_i,k_i}\left[\mathbf{1}\left(\nu_i \geq \hat{\nu}^{\text{rs}}\right) q_i^{\text{rs}}\right], \tag{30}$$

and the seller chooses p to maximize $\pi^{\text{rs}}(p) = (p - \theta^c)D^{\text{rs}}$, with $S_i^{\text{rs}} = CS_i^{\text{rs}} + (p - \theta^c)$.

D Scrap-Value Sensitivity

The main text varies scrap to $0.65\,p_t$ and $0.85\,p_t$. The figures below repeat the normalized policy decomposition and the partial-RPD ϕ grid.

D.1 Scrap value 0.65 p_t

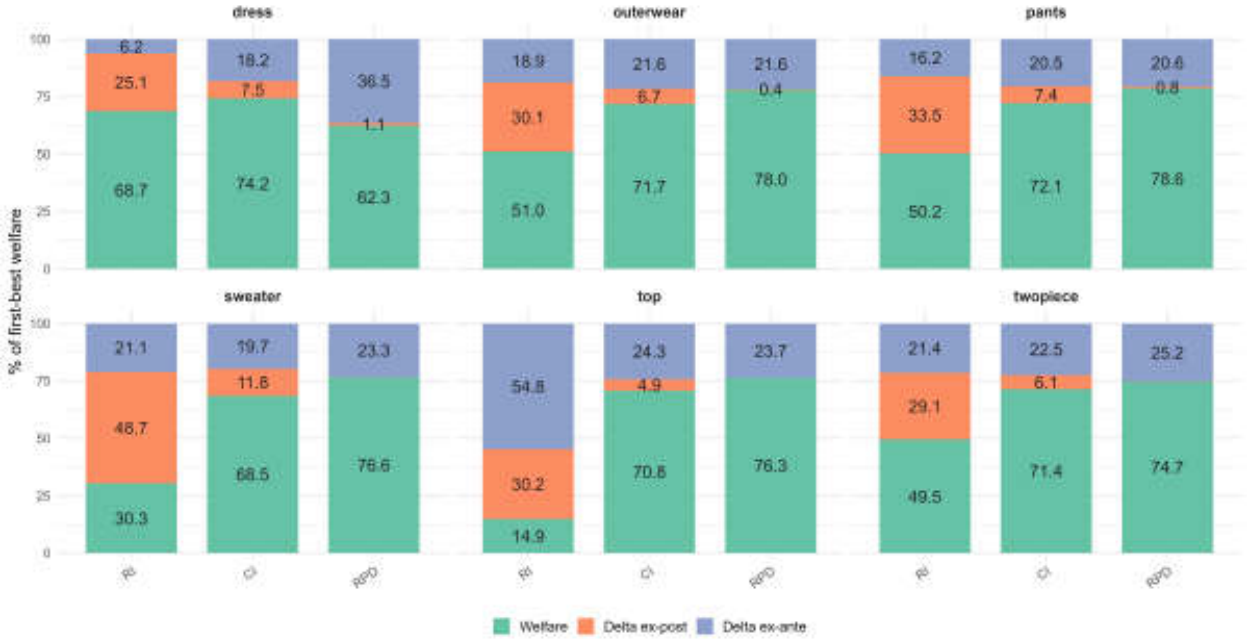


Figure 9: Welfare decomposition as a share of first best. Scrap 0.65 p_t.

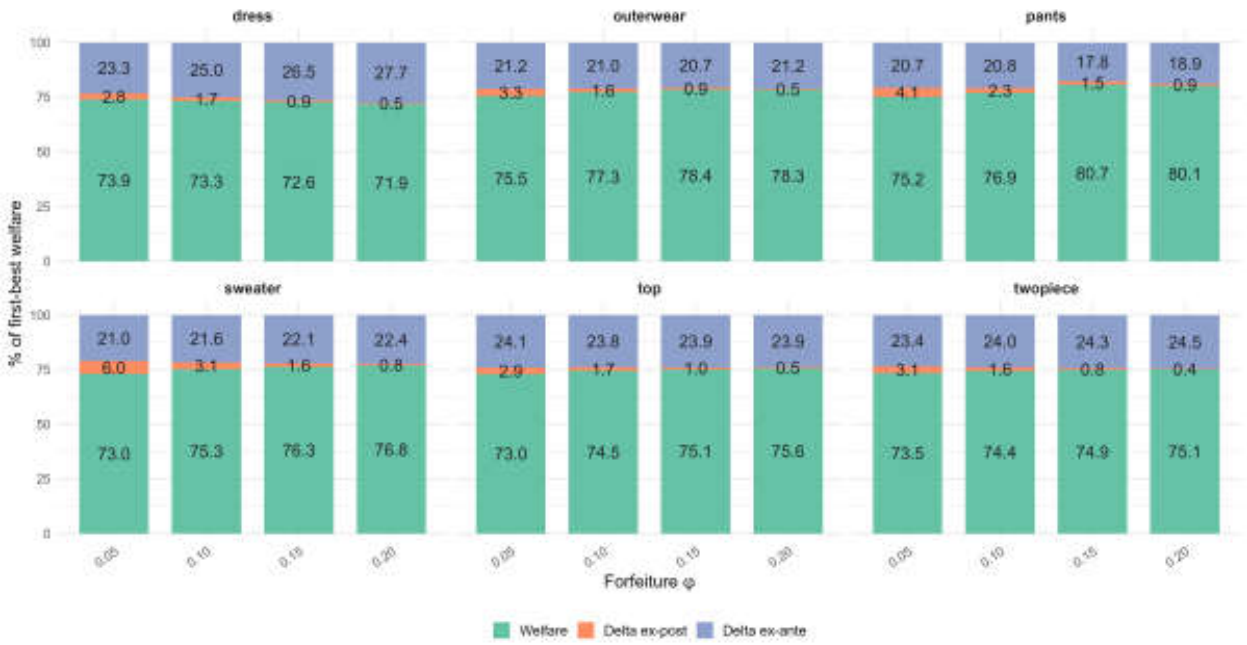


Figure 10: Partial-forfeiture RPD by ϕ. Scrap 0.65 p_t.

D.2 Scrap value 0.85 p_t

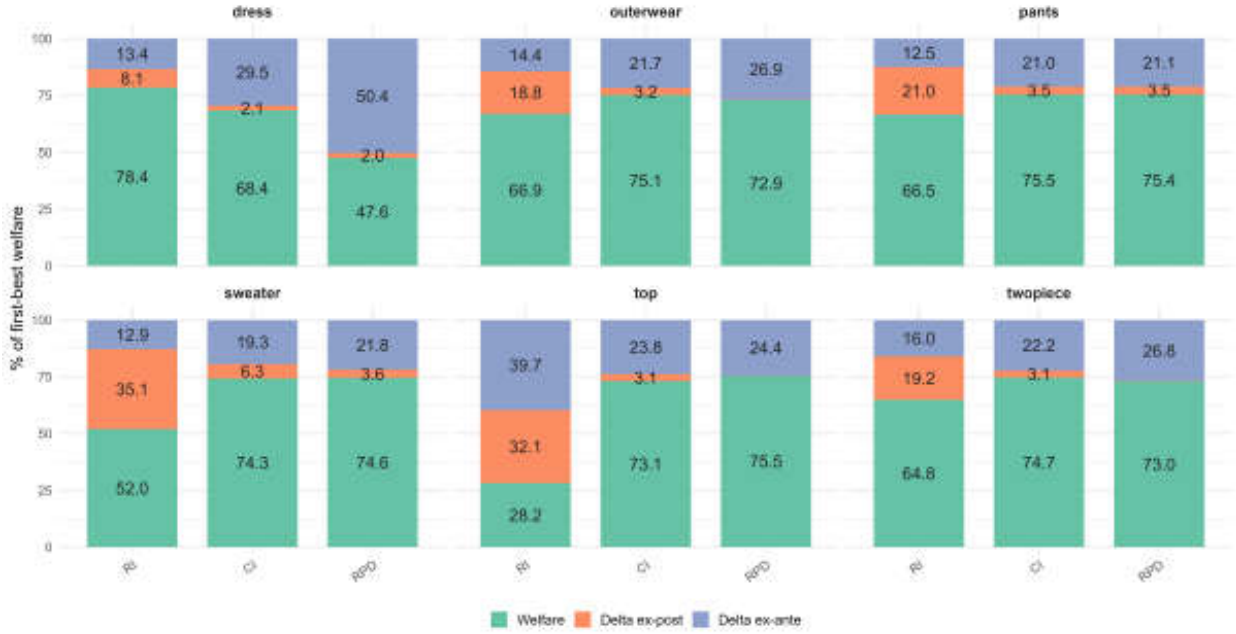


Figure 11: Welfare decomposition as a share of first best. Scrap 0.85 p_t.

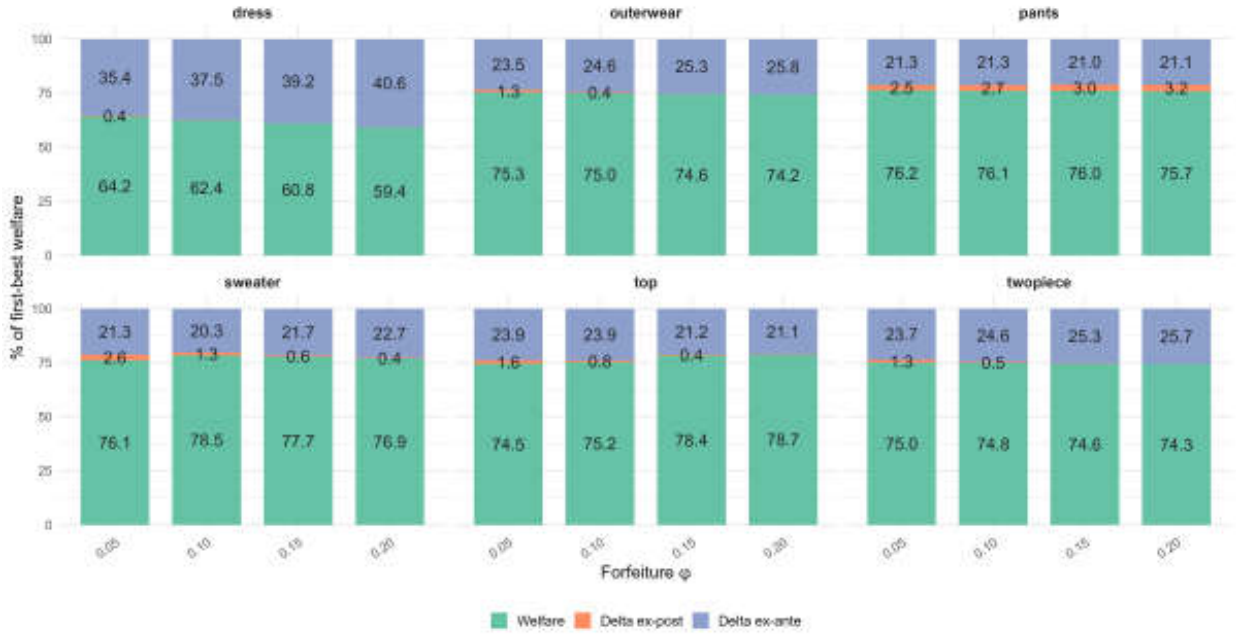


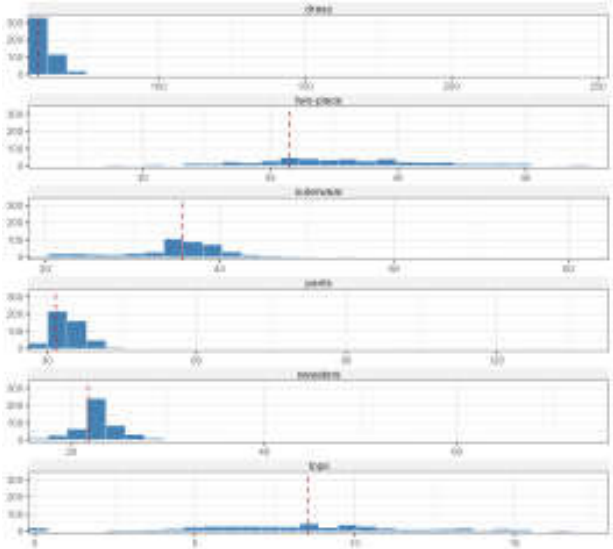
Figure 12: Partial-forfeiture RPD by ϕ. Scrap 0.85 p_t.

E Bootstrap Inference

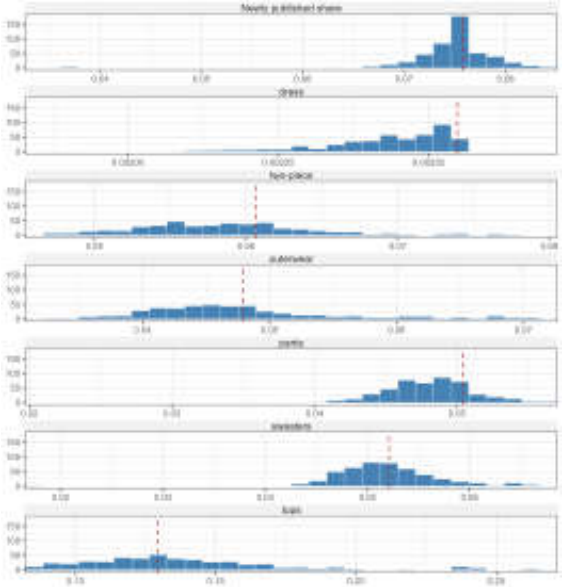
Each panel plots bootstrap replicates in 30 bins. Axes span the 2.5th–97.5th percentile of the displayed draws (by coefficient for parameters; pooled across policies within a category for welfare, scaled to first best). The dashed line is the full-sample point estimate. Main-text standard errors use all replicates.

E.1 Point estimates

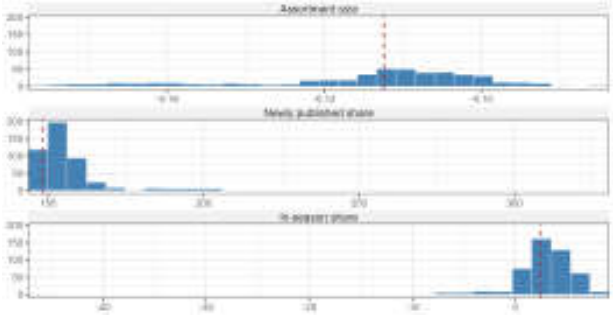
Ex-ante uncertainty (β^{ω})



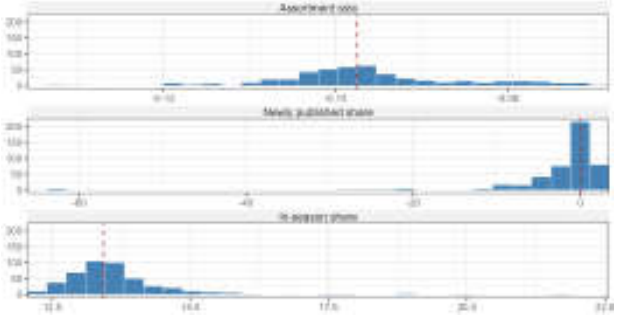
Ex-post uncertainty (β^{σ})



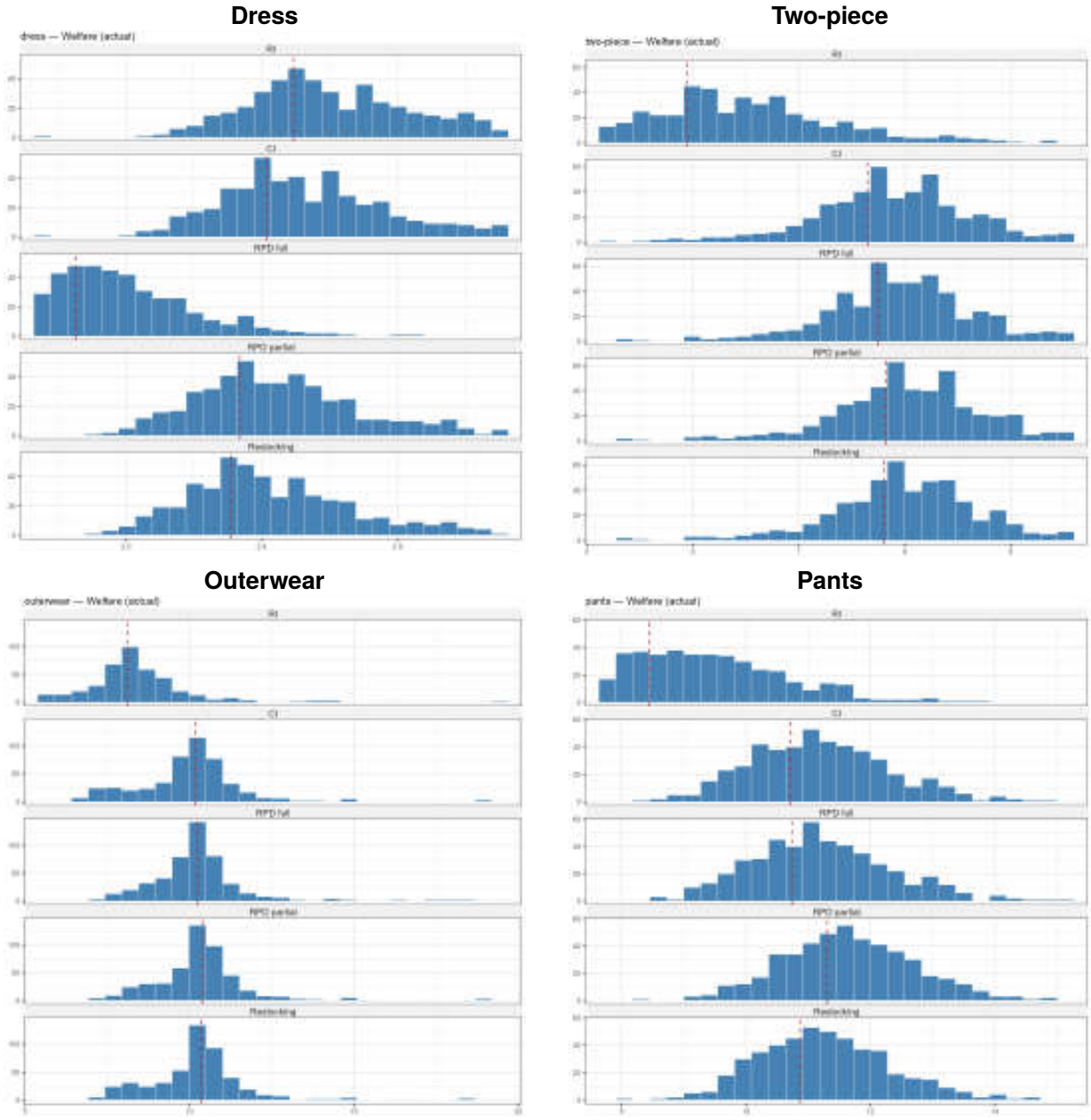
Mean utility (β^{μ})



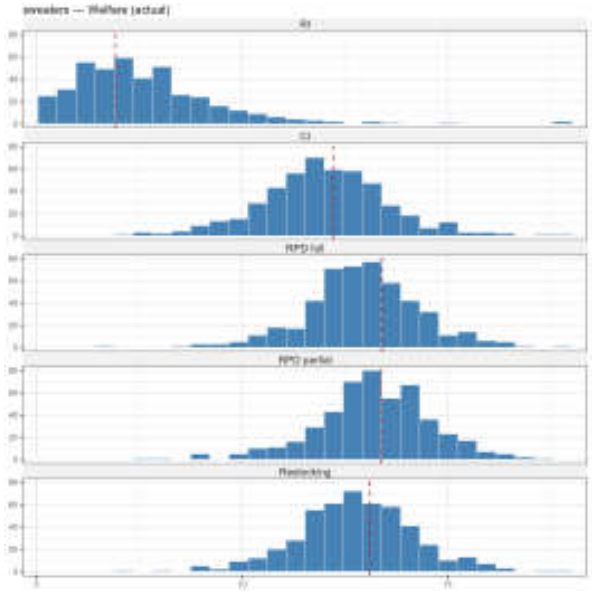
Production cost (θ^{c})



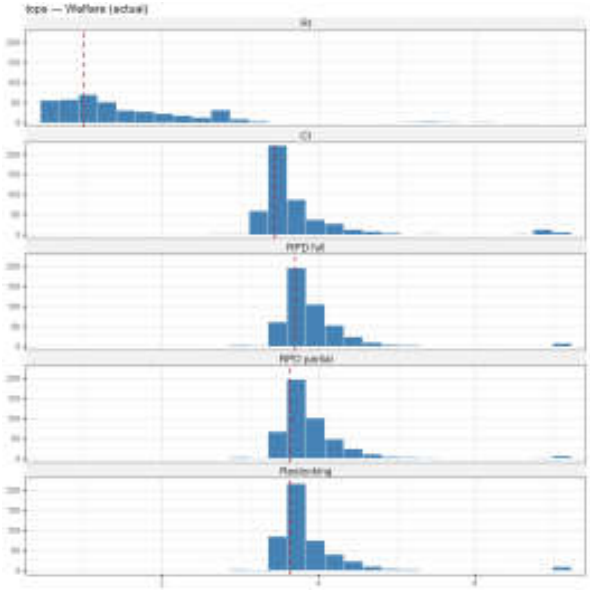
E.2 Actual welfare



Sweaters

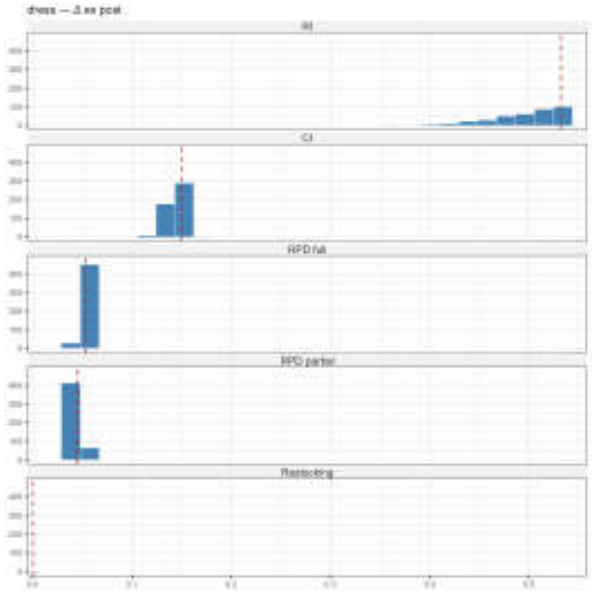


Tops

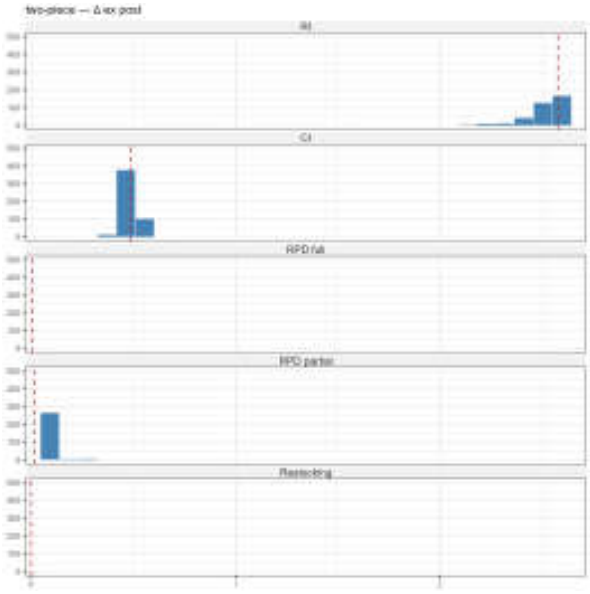


E.3 Δ ex-post

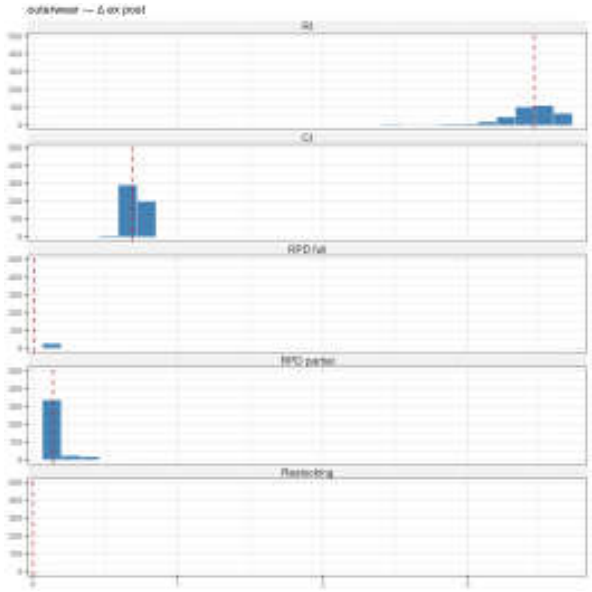
Dress



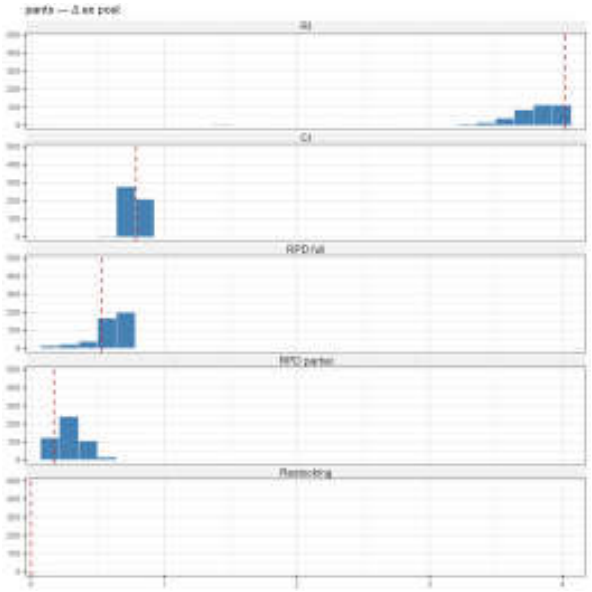
Two-piece



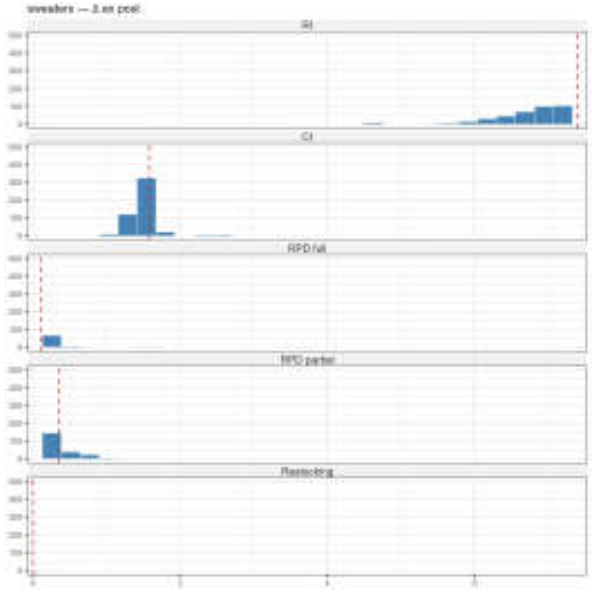
Outerwear



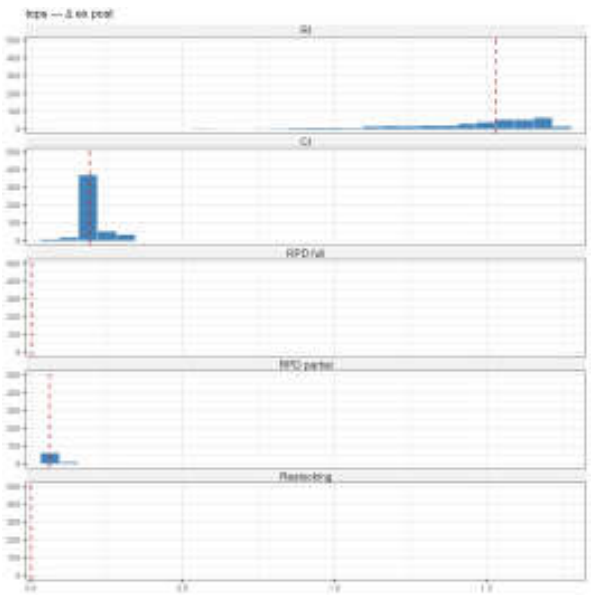
Pants



Sweaters

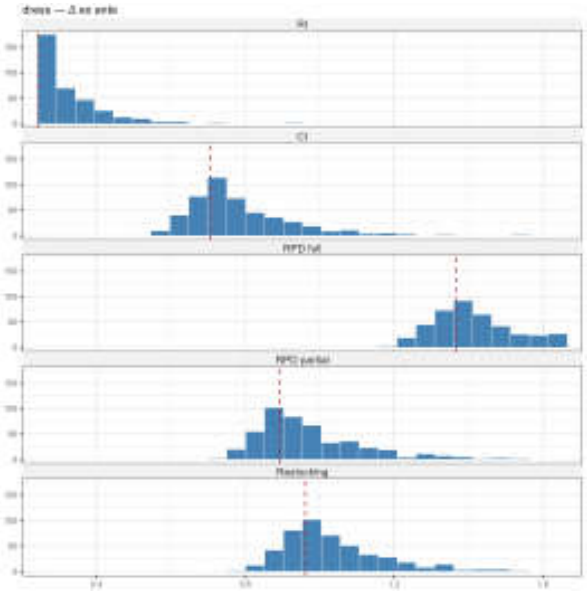


Tops

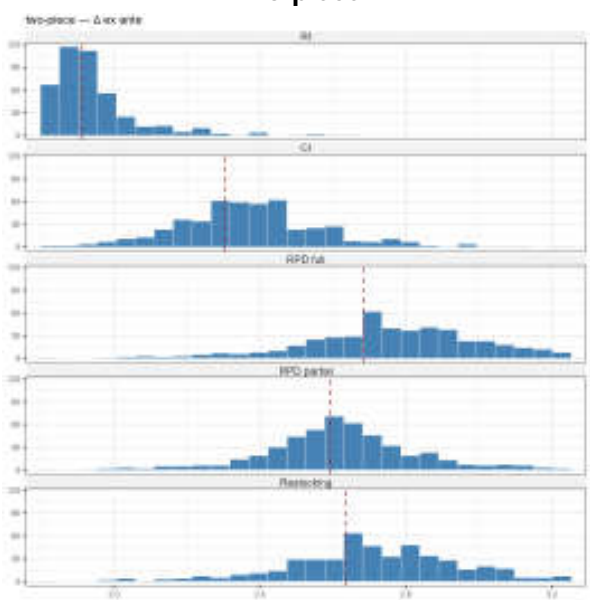


E.4 Δ ex-ante

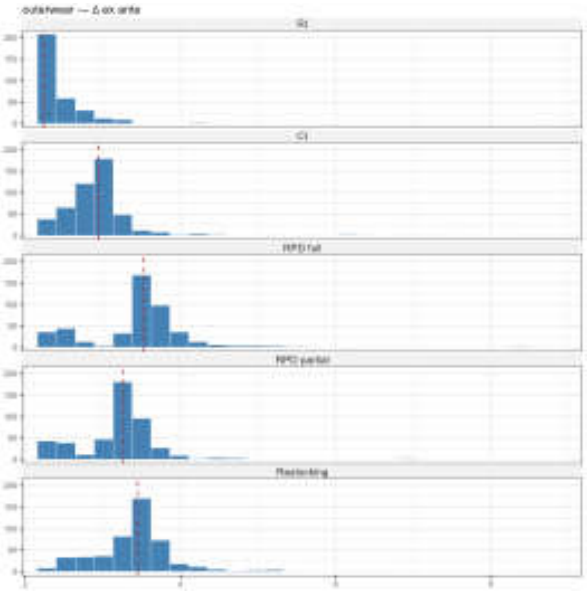
Dress



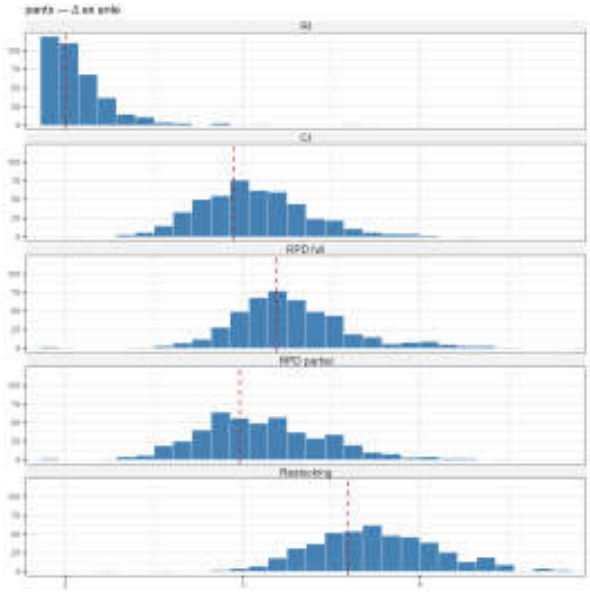
Two-piece



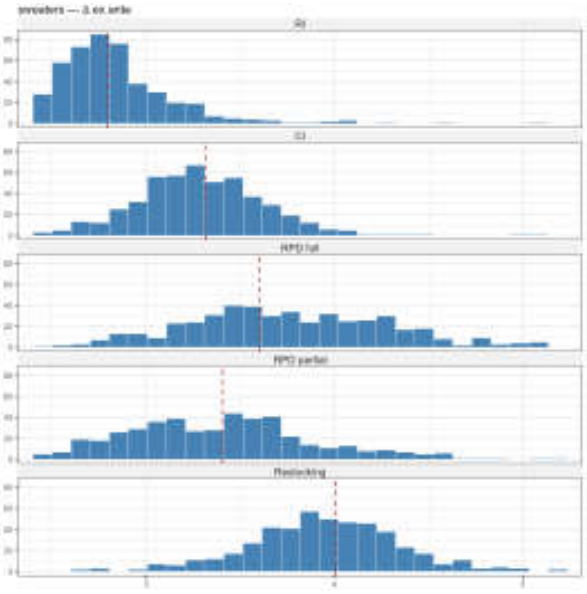
Outerwear



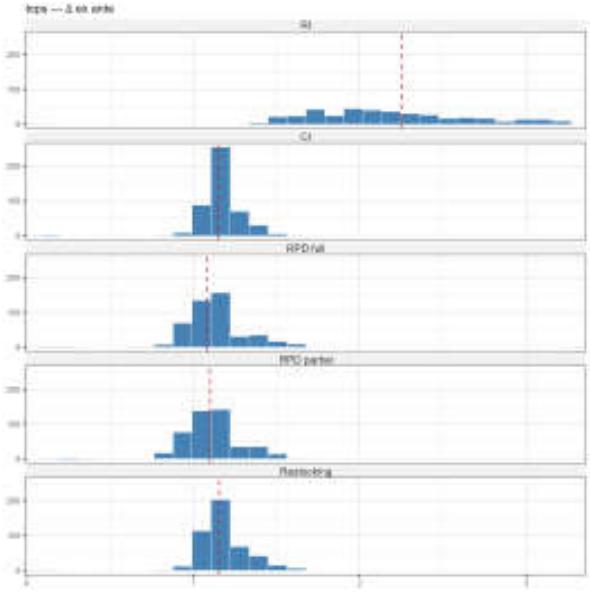
Pants



Sweaters



Tops



E.5 Pairwise win rates

Table 8: Pairwise bootstrap win rates for actual welfare.

Dress

	RI	CI	RPD full	RPD partial	Restocking
RI	—	1.00	1.00	1.00	1.00
CI	0.00	—	1.00	1.00	1.00
RPD full	0.00	0.00	—	0.00	0.00
RPD partial	0.00	0.00	1.00	—	0.89
Restocking	0.00	0.00	1.00	0.11	—

Two-piece

	RI	CI	RPD full	RPD partial	Restocking
RI	—	0.00	0.00	0.00	0.00
CI	1.00	—	0.25	0.00	0.05
RPD full	1.00	0.75	—	0.00	0.00
RPD partial	1.00	1.00	1.00	—	1.00
Restocking	1.00	0.95	1.00	0.00	—

Outerwear

	RI	CI	RPD full	RPD partial	Restocking
RI	—	0.04	0.04	0.04	0.04
CI	0.96	—	0.30	0.05	0.10
RPD full	0.96	0.70	—	0.07	0.20
RPD partial	0.96	0.95	0.93	—	0.99
Restocking	0.96	0.90	0.80	0.01	—

Pants

	RI	CI	RPD full	RPD partial	Restocking
RI	—	0.02	0.02	0.02	0.00
CI	0.98	—	0.24	0.02	0.21
RPD full	0.98	0.76	—	0.01	0.25
RPD partial	0.98	0.98	0.99	—	0.98
Restocking	1.00	0.79	0.75	0.02	—

Sweaters

	RI	CI	RPD full	RPD partial	Restocking
RI	—	0.02	0.04	0.03	0.03
CI	0.98	—	0.04	0.03	0.04
RPD full	0.96	0.96	—	0.15	0.64
RPD partial	0.97	0.97	0.85	—	0.99
Restocking	0.97	0.96	0.36	0.01	—

Tops

	RI	CI	RPD full	RPD partial	Restocking
RI	—	0.00	0.00	0.00	0.00
CI	1.00	—	0.00	0.00	0.00
RPD full	1.00	1.00	—	0.62	0.50
RPD partial	1.00	1.00	0.38	—	0.56
Restocking	1.00	1.00	0.50	0.44	—

Notes: Each off-diagonal cell is the share of bootstrap draws in which the row policy has higher actual welfare than the column policy.